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A Toolbox Approach to Improving the Measurement of Attention Control

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Abstract

Cognitive tasks that produce reliable and robust effects at the group level often fail to yield reliable and valid individual differences. An ongoing debate among attention researchers is whether conflict resolution mechanisms are task-specific or domain-general, and the lack of correlation between most attention measures seems to favor the view that attention control is not a unitary concept. We have argued that the use of difference scores, particularly in reaction time, is the primary cause of null and conflicting results at the individual differences level, and that methodological issues with existing tasks preclude making strong theoretical conclusions. The present article is an empirical test of this view in which we used a toolbox approach to develop and validate new tasks hypothesized to reflect attention processes. Here, we administered existing, modified, and new attention tasks to over 400 subjects (final $N = 396$). Compared to the traditional Stroop and flanker tasks, performance on the accuracy-based measures was more reliable, had stronger inter-correlations, formed a more coherent latent factor, and had stronger associations to measures of working memory capacity and fluid intelligence. Further, attention control fully accounted for the relationship between working memory capacity and fluid intelligence. These results show that accuracy-based tasks can be better suited to individual differences investigations than traditional reaction time tasks, particularly when the goal is to maximize prediction. We conclude that attention control is a unitary concept.

Keywords: individual differences, attention control, measurement

Building new tools to assess individual differences in attention control

Executive attention is studied across many disciplines of psychology and plays a central role in most models of higher-order cognition (Atkinson & Shiffrin, 1968; Baddeley & Hitch, 1974; Botvinick et al., 2001; Egeth & Yantis 1997; Norman & Shallice, 1986; Posner & DiGirolamo, 1998; Shipstead, Harrison, & Engle, 2016). Broadly defined, executive attention guides the control of thoughts and behavior with intention in a goal-driven manner and is particularly important when there is conflict between more automatic processes and one's intentions. It has been shown that individual differences in the ability of executive attention, which we will refer to as attention control, predict higher-order cognitive abilities (Engle, 2002) and is important for many every-day behaviors, including self-control (Broadway, Redick, & Engle, 2010), emotional regulation (Schmeichel & Demaree, 2010), and task-engagement (Miller & Cohen, 2001; Botvinick, Cohen, & Carter, 2004). Still, there remain many theoretical questions as to the nature of attention control and its relation to other cognitive abilities and every-day behaviors. For instance, a theoretical question relevant to our own research is whether a domain-general factor of attention control is the basis of individual differences in higher-order cognitive abilities such as working memory capacity and fluid intelligence (Shipstead et al., 2016).

Unfortunately, such theoretical questions may need to be put on hold until a more fundamental issue that has long plagued individual differences research on executive attention is resolved (e.g., Friedman & Miyake, 2004). The issue is that many executive functioning measures have poor psychometric properties, and it is only recently that this problem has garnered wide-spread recognition (see Draheim, Mashburn, Martin, & Engle, 2019; Hedge, Powell, & Sumner, 2018; Paap & Sawi, 2016; Rouder & Haaf, 2019; Rouder, Kumar, & Haaf,

2019). This has become a contentious topic in the field of attention control. Problems with existing measures calls into question the capability of researchers to measure individual differences in attentional abilities, which in turn casts doubt on conclusions made from previous research and renders it difficult to confidently test theoretical hypotheses related to the domain-generalizability of attention.

Current Views on the Measurement and Assessment of Attention Control

Substantive vs. methodological debate

There are competing views for why robust experimental tasks of attention control do not produce reliable and valid individual differences. One argument focuses on the methodological issues associated with the tasks. Another argument is that the lack of shared variance among most attention measures is more substantive and reflects the domain-specific nature of what these tasks are thought to measure; that is, there is no domain-general attention control ability that can explain performance across different attention-related tasks (Hedge, Powell, Bompas, & Sumner, 2020; Rey-Mermet et al., 2018; Rouder & Haaf, 2019).

On the methodological side, there are known psychometric issues with using well-established experimental paradigms for correlational research. Studies of individual differences in cognition tend to rely on tasks which come directly out of the experimental tradition. The advantage of this is that experimental tasks are based upon a large body of experimental findings and ostensibly do a good job of isolating specific cognitive processes of interest (see Rouder & Haaf, 2019; but see Miller & Ulrich, 2013; Rey-Mermet, Gade, Souza, von Bastian, & Oberauer, 2019; and Verhaeghen & De Meersman, 1998 for challenges to this view). However, tasks which produce robust and reliable effects at the experimental level are often unreliable and weakly correlated with other theoretically related tasks at the individual differences level. Within the

domain of executive functioning, this issue is particularly salient in the study of attention control (e.g., Friedman & Miyake, 2004; Goodhew & Edwards, 2019; Hedge et al., 2018; Magnusdottir et al., 2019; Rey-Mermet et al., 2018; 2019; Paap & Sawi, 2016; Rouder & Haaf, 2019; Rouder et al., 2019), though certainly this is a concern in behavioral science as a whole (see Draheim, Mashburn et al., 2019; Rouder et al., 2019 made a similar observation). The end result is that the integration of experimental tasks for correlational purposes is not as straightforward as one might assume (see Fisher, Medaglia, & Jeronimus; Goodhew & Edwards, 2019; Hedge et al., 2018; and Logie, Della Sala, Laiacona, Chalmers, & Wynn, 1996).

One likely explanation of why experimental tasks are poorly suited to individual differences research is because experimental tasks are designed to minimize between-subject variability and be most sensitive to differences between experimental conditions (Hedge et al., 2018). This is ideal for group comparisons, but the presence of between-subject variability is critical to correlational studies. We argue that difference scores contribute significantly to this problem (for a review, see Draheim, Mashburn et al., 2019). Difference scores are ideal for experimental research as a way to control for baseline performance. The problem for individual differences research is that a difference score is necessarily less reliable than its components¹, resulting in less between-subject variance and therefore attenuated correlations. Difference scores that are low in reliability can counterintuitively result in an *increase* in power in ANOVA-based tests so long as the component scores are reliable (Chiou & Spreng, 1996; Overall & Woodward, 1975). In other words, maximizing power in testing for group differences and maximizing power (reliability) in assessing individual differences are at odds with each other.

Difference scores are therefore useful and perhaps even necessary for isolating specific cognitive

¹ This is assuming that the components are not perfectly reliable or completely independent, which is generally a safe assumption in behavioral research (see Cronbach & Furby, 1970; Lord, 1963).

processes within an individual and finding effects in *experimental* research, but they are not as useful or necessary for isolating cognitive processes between individuals and often demonstrate low reliability and validity in *correlational* research. Many researchers therefore caution against the use of difference scores, particularly in correlational pursuits (Cronbach & Furby, 1970; Draheim, Hicks, & Engle, 2016; Draheim, Mashburn et al., 2019; Edwards, 2001; Goodhew & Edwards, 2019; Hedge et al., 2018; Hughes et al., 2014; Lord, 1956; 1963; Paap & Sawi, 2016).

A related methodological explanation for the poor psychometric properties of attention tasks is that there is too much measurement error due to trial-level variability in performance within individuals (Rouder & Haaf, 2019; Rouder et al., 2019). Specifically, Rouder and colleagues argue that the ratio of trial-level noise to true individual variation is too large in part due to small effect sizes in the difference scores of tasks such as Stroop, flanker, and Simon. Rouder et al. defended the contrast (difference score) approach to isolating variance of interest but also concluded that it might not be possible to increase the ratio of trial noise to true score variance in existing inhibition tasks without *substantially* increasing the number of trials administered (possibly requiring over 1,000 trials!). Therefore, it would appear that the sheer number of trials necessary to provide favorable conditions for the use of difference scores in correlational research effectively precludes their use, especially in studies that involve more than a handful of tasks.

Yet another methodological argument is that performance in the commonly-used Stroop, flanker, and Simon tasks reflect little variance associated with the conflict-resolution processes they are meant to measure, and are overly contaminated with construct-irrelevant variance such as processing speed and speed-accuracy tradeoffs (e.g., Hedge et al., 2020). We discuss this idea more in the following section.

On the other hand, there are also researchers who argue that attention measures do not correlate due to theoretically meaningful reasons. Such researchers may recognize various methodological shortcomings with existing attention measures, but argue that the issues are overblown and/or that null or conflicting results are due primarily to the nature of the constructs in question. Perhaps attention measures do not correlate because attention control² mechanisms are not domain-general and therefore variance in attention control tasks is highly task-specific. For instance, Paap and Sawi (2014) wrote in regard to the weak correlations among attention tasks: "...measures of inhibitory control derived from the flanker, Simon, and Stroop tasks have often shown low levels of convergent validity making it highly likely that the conflict resolution mechanisms employed are task specific rather than recruiting general-purpose inhibitory control." Similarly, although Rouder and Haaf (2019) acknowledged several methodological problems with the assessment of attention control/inhibition, their failure to find a meaningful relationship between Stroop and flanker performance using a hierarchical regression approach designed to account for some of these problems led them to conclude that inhibitory mechanisms in Stroop and flanker are task-specific.

Rey-Mermet et al. (2018) reached a similar conclusion after testing a wide array of commonly used attention control measures in a correlational study with a diverse sample of younger and older adults. Performance on their tasks was scored primarily as reaction time difference scores, and four of their eleven attention measures were standard Stroop and flanker tasks. Only 25% of their task-level correlations were statistically significant, only 11% of the correlations exceeded $r = .20$, and some correlations were in the opposite direction such that better performance on one attention task was associated with worse performance in another.

² What we refer to as attention control is sometimes called inhibition by other researchers. For the present purposes, there is no meaningful difference between these two terms.

When all tasks were loaded onto a single attention factor, half had factor loadings below .20, which is far from acceptable even if one adopts a liberal tolerance (e.g., Comrey & Lee, 1992; Matsunaga, 2010; Stevens, 1992). They interpreted their results as demonstrating the lack of a unified inhibition factor. However, we would argue that their findings only reinforce the poor psychometric properties of these tasks). Only two of their difference score measures had a corrected split-half internal consistency at or above .75, with an average of .64.³ In a more recent study, Rey-Mermet et al. (2019) administered accuracy-based tasks using a novel calibration procedure but still failed to find a unitary attention control factor. We provide a commentary of their study toward the end of the discussion section with potential reasons for their null results.

Our view is that the weak correlations across various attention control tasks is methodological in nature. We argued in Draheim, Mashburn et al. (2019) that the use of reaction time, especially difference scores in reaction time, is one of the primary reasons for issues with tasks such as Stroop and flanker.⁴ In addition to the poor psychometric properties of difference scores, reaction time measures in general are sensitive to speed-accuracy interactions which can manifest in a number of ways (for reviews, see Draheim, Mashburn et al., 2019; Heitz, 2014), and these are problematic for differential and developmental studies. No studies in support of the domain-specific view of attention control have appropriately addressed the methodological issues of using reaction time difference scores, or difference scores more generally. Further, this view relies on the acceptance of the null hypothesis, that non-significant correlations between related tasks reflects the absence of common variance. This is particularly problematic in

³ Although rules of thumb and guidelines are by no means absolute, .80 is often quoted as the threshold for acceptable reliability in basic research (e.g., Nunnally, 1964).

⁴ Accuracy-based difference scores are no better than reaction time difference scores (e.g., Hughes et al., 2014; Rey-Mermet et al., 2019), but they are less commonly used in measuring executive functioning, and so it is the reliance upon reaction time difference scores which is our focus here.

individual differences research in which so many factors can unknowingly contribute to null findings.

Reaction time is contaminated with speed-accuracy tradeoffs and processing speed

Many reaction time tasks are scored without any consideration or control for accuracy. This is an issue because subjects who have quick and error-prone responding will appear better on the task than subjects who have more deliberate and slower responding but with fewer errors solely due to differences in emphasis on speed and accuracy as opposed to differences in levels of the ability of interest. Difference scores are used to account for baseline processing (task fluency and processing speed) but a difference in reaction times does not account for differences in speed-accuracy tendencies. Another consideration is that researchers have questioned the assumption that reaction time difference scores wholly control for outside sources of variance, namely those related to general processing speed or ability. Miller and Ulrich (2013) argued that the cognitive processes underlying reaction times is not as simple as often believed and that interpretations predicated on correlations between reaction times are often faulty. The crux of their argument is that reaction times are impure, and that observed correlations involving reaction time measures have multiple influences, some of which map to processes of interest whereas others do not. It has also been established that reaction time differences do not properly account for processing speed (e.g., Rey-Mermet et al., 2019; Verhaeghen & De Meersman, 1998). Rey-Mermet et al. (2019) pointed out there is a confound between processing speed and cognitive ability in tasks in which performance is scored as a reaction time difference score. Ergo, even if reaction time differences in attention control tasks (such as Stroop and flanker) revealed large individual differences that correlated to performance on other tasks, it would not be clear whether these individual differences reflected differences in attentional abilities or

merely differences in general processing speed. To that end, Hedge et al. (2020) showed that, although most attention control tasks weakly correlate with one another, processing speed and speed-accuracy tradeoffs are shared across these tasks. Specifically, they used diffusion modeling to reanalyze several datasets which used a combination of Stroop, flanker, and Simon tasks and found that despite low correlations of performance data (error and reaction time costs), non-decision time and variability, drift rate (associated with processing speed), and boundary separation (associated with response cautiousness and therefore speed-accuracy tradeoffs) parameters were each strongly correlated within and across tasks. Diffusion parameters associated with conflict resolution (which these tasks were designed to measure) were not correlated within or across tasks. They concluded that these conflict tasks are contaminated with sources of construct irrelevant variance and therefore that a hypothetical positive relationship among them would be only minimally informative due to this contamination. These findings separately support the Rey-Mermet et al. (2019) concern that reaction time interference effects in these conflict tasks may be contaminated with processing speed, and our argument that speed-accuracy tradeoffs and interactions are a concern with executive functioning tasks (Draheim et al., 2016; Draheim, Mashburn et al., 2019). Hedge et al. also performed a simulation on these tasks and showed that even if the diffusion parameters associated with conflict were very strongly correlated across tasks, this would not necessarily result in strong correlations in the behavioral data. Hedge et al.'s findings call into question the widely held belief that difference scores in conflict tasks such as Stroop, flanker, and Simon are theoretically meaningful.

Although Hedge et al. (2020) found that contamination from processing speed and speed-accuracy tradeoffs was present in both reaction time and accuracy difference scores, it should be noted that the Stroop, flanker, and Simon tasks were designed in such a way that both reaction

time and accuracy are important to performance. That is, respondents must balance the two and decide how much emphasis to give to both while performing the task. We argue that it is easier to account for extraneous sources of individual differences variance such as processing speed and speed-accuracy interactions when using accuracy-based measures. Whereas it is difficult to assess executive functioning with a reaction time task in which accuracy is completely irrelevant, it is much more straightforward to design an accuracy-based executive functioning task in which reaction time is rendered irrelevant. A subject can respond as quickly or as slowly as they want, provided they are not responding too quickly to make motor errors or too slowly such that the memory representation of the trial is lost. If our reasoning is correct, this is quite a wide time window to respond without appreciably affecting performance. Wickelgren (1977) made a very similar point in reference to ways to account for speed-accuracy tradeoffs in cognitive tasks:

Although the basic fact of speed-accuracy tradeoff may make it inadequate to look at reaction time alone as the dependent variable, it does not invalidate looking at asymptotic accuracy as a dependent variable without reference to reaction time. The fact that speed-accuracy tradeoff functions approach an asymptotic level of accuracy at long reaction times means that so long as the response time is sufficiently long to ensure that one is operating at or near the asymptote, enormous differences in reaction time will be associated with negligible differences in asymptotic accuracy. Thus, **there is little opportunity for contamination of asymptotic accuracy by differences in response time, while there is considerable opportunity for contamination of reaction time by differences in accuracy.** (p. 82; emphasis and underline added)

Reaction-time irrelevant tasks may be a solution to the measurement problem

This observation by Wicklegren (1977) implies something rather intuitive. Because speed-accuracy tradeoffs are a concern in cognitive tasks, one way to account for them is to render reaction time irrelevant to the task such that respondents do not have to emphasize one over the other. In other words, one way to account for speed-accuracy tradeoffs is to use tasks which apply relatively little demand to respond quickly and/or otherwise make quick responding unnecessary or impossible. If this is accomplished, then the variance of interest should all be in accuracy.

The antisaccade task is an example of one measure in which reaction time is irrelevant, and therefore variance of interest is entirely in accuracy. On this task, subjects must make a saccade in the opposite direction (left or right) of a cued stimulus and identify a target stimulus that is quickly masked. If they do not make the antisaccade in time, then they will miss the target stimulus. They have as long as they want to respond, however they either saw the target stimulus or they did not and therefore their response time is irrelevant. We argue that construct irrelevant variance such as that from processing speed and speed-accuracy tradeoffs are therefore minimal in this task. This could explain Rey-Mermet et al.'s (2019) observations that antisaccade tasks tend to “dominate” latent factors of attention. Our lab similarly finds that loadings for the antisaccade task tend to be much higher than the .20 - .40 observed with traditional Stroop and flanker tasks, which fall below acceptable levels and indicate that our so-called attention factor is comprised mostly of variance from a single task – the antisaccade (e.g., Shipstead, Harrison, & Engle, 2015). Relatedly, we find the antisaccade to be highly reliable (around .90) and the Stroop and flanker tasks to be much less so (.60 - .70), and that correlations involving the antisaccade task and other cognitive measures are much stronger (as strong as $r = .50$) whereas correlations

involving Stroop and flanker are numerically half at best, and our Stroop and flanker tasks share only 3 - 4% common variance at the task level (around $r = .10$).

Given that recent attempts to improve the measurement of attention control (e.g., Rey-Mermet et al., 2019; Rouder et al., 2019) have not been successful (but see Paap, Anders-Jefferson, Mikulinsky, Masuda, & Mason, 2019 for one report of strong correlations with Stroop and Simon tasks using an integrative speed-accuracy score). We believe that the most practical approach to developing improved measures of attention control is to create tasks that are reaction time irrelevant, accuracy-based, and avoid the use of difference scores. This research endeavor is somewhat exploratory in the sense of task-development, however, is hypothesis-driven because it provides a test of the methodological vs. substantive debate as to why common variance is not shared across attention control tasks. Further, we can also ask more specific theoretical questions such as whether a unitary factor of attention control can fully account for the working memory capacity-fluid intelligence relationship.

Goals of the Present Study

The present study was an effort to test whether developing new and modified attention tasks would lead to improvements in the ability to measure individual differences in attention control. Given the issues of reaction time and difference scores, we reasoned that pushing performance variance away from reaction time and into accuracy would be a viable approach for improving the psychometric properties of attention tasks. Accuracy-based and adaptive procedures can circumvent the noted issues with difference scores and allow for better control of speed-accuracy interactions and processing speed. Due to the aforementioned concerns regarding contamination from processing speed and other construct-irrelevant sources of variance (Hedge et al., 2020), we also included measures of processing speed to assess discriminant validity.

Formally stated, the primary question of the present study was whether this toolbox approach to developing new varieties of attention control tasks would result in improvements in the measurement of attention control. The evaluative criteria used to assess this question were, 1) task reliability, 2) inter-correlations with other attention control tasks, 3) latent factor coherence, 4) the relationship to working memory capacity and fluid intelligence, and, 5) the mediation of the working memory capacity/fluid intelligence relationship. Reliability is the first criterion since a measure cannot be valid without being reliable, and the unreliability of many attention tasks (such as the reaction time difference score-based Stroop and flanker) has been called into question by us and other researchers. Second, an attention control measure should correlate strongly to other attention control measures if indeed, as we argue, attention control is a broad, unified, and domain-general construct. Third, another goal for improving the measurement of attention control, if attention control is a unified domain-general ability, should be to find tasks which have a more balanced loading onto a latent factor. Finally, according to our theoretical view of executive attention, attention control measures should strongly relate to working memory capacity and fluid intelligence and attention control should fully mediate the relationship between working memory capacity and fluid intelligence. Predicting real-world behavior is an important criterion as well, however we address this topic in a separate paper (Martin, Mashburn, et al. 2019) which focuses on using accuracy-based attention control measures to predict incremental variance in multitasking ability above and beyond the Armed Services Vocational Aptitude Battery (ASVAB).

The Toolbox of Attention Control Measures

We compared several different types of attention control measures. The first group consisted of existing measures; the reaction time difference score-based color Stroop, the

reaction time difference score-based arrow flanker; error rates in a non-adaptive accuracy-based antisaccade task; a reaction time-based psychomotor vigilance task; and a capacity-based visual arrays (change detection) task requiring selection/filtering of some elements of the display. The first three are tasks our lab has traditionally used to assess attention control. Although the standard Stroop and flanker tasks are known to be poor individual differences measures, they were included for comparative reasons. The psychomotor vigilance task is a reaction time measure, but it may be an example of one well suited to correlational analysis because it is not scored as a difference and, presumably, speed-accuracy interactions would not be present (there are no right or wrong answers once the counter begins ticking up, only quicker and slower responses). So, just as tasks such as the complex span and antisaccade effectively control for speed (response time is irrelevant), the psychomotor vigilance task effectively controls for accuracy (an inaccurate response is not possible). As for the inclusion of visual arrays as an attention measure, see the following section.

The second group of tasks were threshold-based modified Stroop and flanker tasks that employed an adaptive procedure. We reasoned that modifying these tasks to be adaptive threshold-based and not at all reliant upon differences in reaction time may be a method of improving the reliability and validity. Further, these tasks were modified in such a way that interference effects due to the intermixing of congruent and incongruent trials should play a role in performance, although said interference was not directly measured with the dependent variable.

The third group of tasks were completely novel tasks. One, sustained attention-to-cue, was designed to be an accuracy analog to the psychomotor vigilance task and with a distractor similar to that of the antisaccade. We reasoned that the requirement for both sustained attention

and resisting a distractor should enhance the amount of controlled attention required to perform the task well. The other, adaptive visual cue, was designed to be similar to antisaccade but adaptive and with a larger number of non-target stimuli. These tasks are available for download at <http://englelab.gatech.edu/attentioncontroltasks>

Selective visual arrays as a measure of attention control

Including the visual arrays as a measure of attention control is controversial and requires explanation. The visual arrays, along with similar change detection paradigms, are generally conceptualized as measures of visual working memory capacity and indeed change detection tasks in general do provide a useful estimate of the amount of information stored in working memory (e.g., Conway, Kane, & Engle; 2003; Luck & Vogel, 1997). However, we argue that there is sufficient evidence that individual differences in visual arrays performance is more so a reflection of attentional abilities than storage or capacity. The question of what the visual arrays measures is an important one and it would not be feasible to fully detail all lines of evidence for our position here, so instead this issue is discussed in full detail in a separate paper which had been submitted for publication and is available to read on PsyArXiv as a pre-print (Martin, Tsukahara et al., 2019).

To highlight, several studies by Vogel and colleagues show the importance of attention in variation in visual arrays performance. For example, Fukuda, Woodman, and Vogel (2015) found that capacity scores in visual arrays was significantly smaller for a supra-capacity set size (8) than for a near-capacity set size (4), which supported the attentional-control account of individual differences in visual arrays capacity scores over the memory storage account. Vogel, McCollough, and Machizawa (2005) found that contralateral delay activity (which increases with the amount of stored information and therefore can be used as an indicator of whether irrelevant

distractors are processed, thereby unnecessarily consuming memory capacity) for low and high working memory capacity individuals was roughly the same for two items and four items arrays in which distractors were not present, but that there was a strong difference in contralateral delay activity in high vs. low spans when two items were presented along with two distractors. This study therefore indicated that lower ability individuals perform worse in visual arrays tasks due to the inability to filter distractor items. Finally, Fukuda and Vogel (2011) showed that high and low spans differed in how long it took them to recover from attentional capture when performing a visual arrays tasks, thus resulting in individual differences in task performance.

Here, a distinction between different types of visual arrays tasks is critical. One type of visual arrays is a more straightforward change detection/primary memory task in which the respondent indicates whether something changed from one display to another (non-selective visual arrays). Another type of visual arrays task places a filtering/selection demand on the respondent such that the respondent must ignore or filter out half of the to-be-presented array (a cue presented very briefly prior to the to-be-remembered array indicates which half of the array the respondent should attend to; selective visual arrays). Although performance in the non-selective visual arrays likely involves a degree of attentional processes (for example Fukuda & Vogel, 2011 and Vogel et al., 2005 used non-selective visual arrays and found that individual differences could be attributed in part to attentional control), we would consider performance in these tasks to be more so driven by working memory capacity than attention control. However, we do consider selective visual arrays to primarily reflect attentional processes and, therefore, to be a measure of attention control. Performance on the visual arrays is assessed using a capacity score – a reflection of how many items the respondent could retain in the array. The capacity scores on versions of the visual arrays without the filtering demand (non-selective visual arrays)

is around 3-3.5 (similar to tasks of working memory capacity), whereas capacity scores on versions with the filtering demand (selective visual arrays) is about half that (e.g., Shipstead et al., 2015), indicating that the filtering demand alters performance in a meaningful way. Further, data from our lab across four separate studies show that selective visual arrays shares more variance with attention control tasks (namely antisaccade) than with working memory tasks (see Martin, Tsukahara et al., 2019; Shipstead, et al., 2015). At the latent level, selective visual arrays accounts for substantial and unique variance in attention control above and beyond working memory capacity tasks and non-selective visual arrays. Additionally, visual arrays with the filtering demand loads onto the same factor as antisaccade and other attention tasks and *not* with measures of working memory capacity. In sum, although selective and non-selective visual arrays tasks share substantial variance and load together latently, they can be separated out as measures which primarily reflect attention control and working memory capacity, respectively.

Method

The study at large consisted of approximately 30 cognitive tasks administered over four two-hour long sessions. Subjects were also invited back for subsequent sessions; a two-hour session to assess test-retest reliability, and two 2-hour long sessions for a separate project from which we present selected data later in the present article. In addition to performance data and demographic information, we also collected eye tracking, pupillometry, and mind wandering data from a number of tasks, which we do not report here.

The data analyzed in this study was part of a larger data collection sample and we reported data that included some of the same tasks in different publications. The following link has a summary of the larger data collection procedure and a reference list of all publications to come out of this data collection sample with information on which tasks were used for each

publication: <https://osf.io/s5kxb>. In the Author's Note we also list the various articles and conference presentations in which some ideas and/or data from the present study were disseminated.

Participants

Subjects completed a battery of cognitive tasks divided into four sessions, each lasting approximately two hours. Subjects could schedule sessions within designated slots during lab hours and with the restriction that they could not complete more than one session on any single day. Subjects received \$130 in compensation, which was distributed as \$25 for the first session with each subsequent session worth \$5 more than the previous. Georgia Tech psychology students could choose to receive either payment or research participation credit for each session (two credits per session). After completing this study, most subjects were invited to return for an additional two sessions for a separate study. After completing that two-session study, they were also invited back for one final session in which we obtained test-retest reliability estimates on the attention control measures. Subjects who frequently rescheduled, failed to show up to appointments, or were otherwise problematic in terms of behavior or performance data (e.g., lack of effort, falling asleep during tasks, failure to understand instructions, or near-floor performance on multiple tasks) were not invited back to the lab for these additional sessions. The experiment room was set up to run up to five subjects at a time with partitions between the kiosks such that subjects could not see each other. Subjects wore headphones while performing each task. An experimenter sat behind them at all times, occasionally observing their performance to ensure subjects were following instructions. The experimenter also took notes regarding subjects' behavior, alertness, and apparent motivation, answered instruction-related questions, and started the run files for each task. Subjects were not told directly that they would be observed but they

were aware of the experimenter's presence. Undergraduate research assistants served as the experimenter the majority of the time, with graduate students or post docs filling in as needed. At least one senior lab member (graduate student or post doc) had to be present in the lab to supervise data collection. Most tasks had built-in rest periods that appeared after a set number of trials for each task, designed to occur approximately every ten minutes. Subjects could advance the rest screen at their convenience to continue performing the task. Subjects were asked to avoid getting out of their chair while in the middle of a task if possible and were encouraged to take short breaks between tasks when needed. The Georgia Institute of Technology Institutional Review Board approved the protocol for this study.

Data collection took place at the Georgia Institute of Technology. A total of 482 subjects enrolled in the study, with 403 finishing all four sessions (60.0% female; 39.5% male; 0.5% other). Just under 94% of subjects reported being a former or current college student (46% Georgia Tech; 22% Georgia State University; 25% other institution). To be eligible for the study, subjects had to report being a native (learned at age 5 or earlier) and fluent English speaker age 18 – 35 years with normal or corrected-to-normal vision, no history of seizure, and having never participated in a previous study in our lab. Subjects were recruited via the Georgia Tech subject pool, flyers around the Georgia Tech campus, flyers off-campus, Reddit and Craigslist postings, various advertising efforts (Facebook, Creative Loafing, campus publications), and from in-person recruiting in the downtown and midtown Atlanta areas.

Tasks

Working Memory Capacity

We measured working memory capacity using three complex span tasks; the advanced operation span, advanced symmetry span and advanced rotation span. The complex span tasks

consist of alternating memory storage and processing sub-tasks (Unsworth et al., 2005). The advanced versions of the tasks included larger set-sizes of memory items (Draheim, Harrison et al., 2018). For each task, to help ensure that subjects were attending to the processing task and not rehearsing the to-be-remembered information during this time, subjects were allotted up to their individual mean reaction time + 2.5 standard deviations from a block of practice trials on the processing task only. Additionally, subjects were asked to maintain a minimum level of performance on the processing tasks (85% accuracy) and their cumulative percent correct was shown to them throughout the task in the upper-left corner of the screen. These tasks can be downloaded at <http://englelab.gatech.edu/taskdownloads.html>.

Operation span (Kane et al., 2004; Turner & Engle, 1989). This task required subjects to remember a series of letters presented in alternation with simple math equations which they were required to solve (Figure 1). On each trial, subjects first solved a simple math equation (e.g., “ $(2 \times 2) + 1 = 5$) or not (e.g., “ $(3 \times 4) - 3 = 8$ ”) followed by the presentation of a single letter. This alternation repeated until a variable set-size of letters to-be-remembered had been presented, at which point subjects attempted to recall the letters they had seen in their correct serial order. There were a total of 14 trials (2 blocks of 7 trials), set-sizes ranged from 3-8⁵, and each set-size occurred twice (once in each block). The dependent variable was the partial span score, which is the total number of letters recalled in proper serial position (Conway et al., 2005).

Symmetry span (Unsworth, Redick, Heitz, Broadway, & Engle, 2009). This task required subjects to remember a series of spatial locations in a 4x4 matrix presented in alternation with a pattern of squares which they had to decide whether the pattern was symmetrical on the vertical

⁵ Due to an error in programming of the advanced operation span task, trials in which the set-size was supposed to be 9 only displayed a set-size of 8. This resulted in the set-size of 8 occurring twice as often as intended, a total of 4 trials compared to 2 trials for each other set-size.

midline (Figure 1). On each trial, subjects were first presented with a 16 x 16 matrix of black and white squares and were required to decide whether the pattern was symmetric on the vertical midline. Followed by the symmetry judgment, a 4 x 4 matrix of squares with one square highlighted in red were displayed. The location of the red-square was the to-be-remembered spatial location. This alternation continued until a variable set-size of spatial locations had been presented. Subjects then attempted to recall the locations of the red square in correct serial order. There were a total of 12 trials (2 blocks of 6 trials), set-sizes ranged from 2-7, and each set-size occurred twice (once in each block). The dependent variable was the partial span score, which is the total number of square locations recalled in proper serial position.

Rotation span (Kane et al., 2004). This task required subjects to remember a series of directional arrows of varying size in alternation with a mental rotation task in which they had to mentally rotate and decide if a letter was mirror reversed or not. On each trial, subjects first solved a mental rotation problem followed by the presentation of a single arrow with a specific direction (8 possible directions; the four cardinal and four ordinal directions) and specific size (small or large). Both the direction and size of the arrow were the to-be-remembered features. This alternation continued until a variable set-size of arrows had been presented, at which point subjects attempted to recall the set of arrows in their correct serial position. There were a total of 12 trials (2 blocks of 6 trials), set-sizes ranged from 2-7, and each set-size occurred twice (once in each block). The dependent variable was the partial span score, which is the total number of arrows recalled in proper serial position.

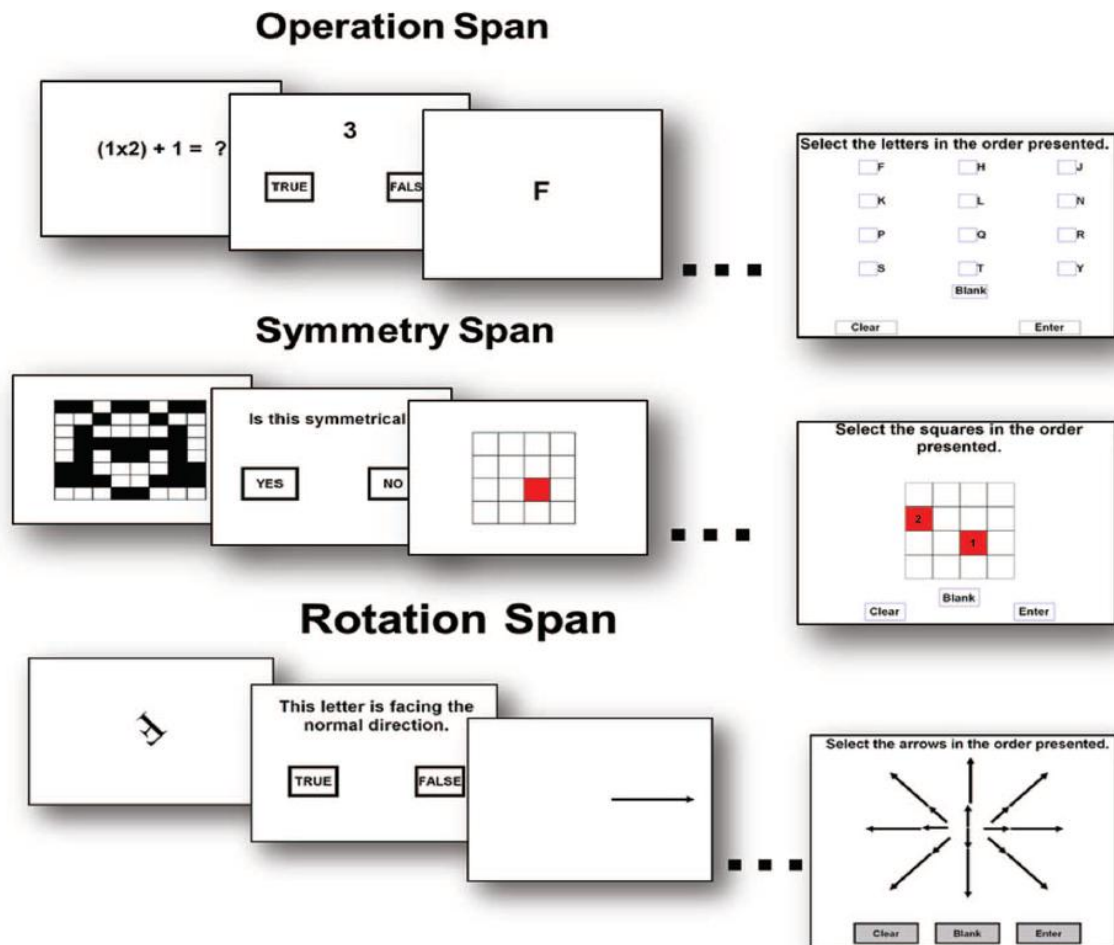


Figure 1 - Demonstration of the complex span tasks. In each task, subjects respond true/false or yes/no to a processing (distractor) task prior to the presentation of each to-be-remembered stimulus. After a variable amount of presentations (depending on the set-size for that trial), a recall screen appears asking the subject to recall the to-be-remember stimulus in order of presentation. The dependent variable is the partial span score, which is the total number of items recalled in the correct position.

Fluid Intelligence

Raven's advanced progressive matrices – Odd problems (Kane et al., 2004; Raven & Court, 1998). Subjects were presented abstract shapes in a 3 x 3 matrix. The shape in the bottom-right was missing, and the subject had to select which item, from the eight answer options, best

completed the overall pattern by clicking that option on the screen. Subjects had 10 minutes to complete 18 problems. The number of correct responses was the dependent variable.

Number series (Unsworth et al., 2009; Thurstone, 1938). Subjects were presented a sequence of numbers and needed to identify the response option that was the next logical number in the sequence by clicking the correct number from five total response options. Subjects had 5 minutes to answer 15 problems. The number of correct responses was the dependent variable.

Letter sets (Ekstrom et al., 1976). On each problem, the subject was presented five sets of letters, each containing four letters that follow a particular rule. Instructions were to find the rule that applied to four of the five letter sets, and then indicate the set that violates the rule by clicking that set on the screen. Subjects had 10 minutes to complete 30 problems. The number of correct responses was the dependent variable.

Non-adaptive Attention Control Tasks

Antisaccade (Hutchison, 2007; Kane et al., 2001). Subjects saw a central fixation cross lasting a random amount of time between 2000 – 3000 ms followed by an alerting tone for 300 ms. After the alerting tone, an asterisk appeared for 300 ms at 12.3° visual angle to the left or the right of the central fixation followed immediately by a target “Q” or an “O” for 100 ms on the *opposite* side of the screen from the asterisk. The location of the asterisk and target letter were both masked for 500 ms. by “###”. The subject’s goal was to ignore the asterisk and instead look away to the other side of the screen to catch the target “Q” or “O”. Subjects had as much time as needed to respond to which letter appeared by pressing the associated key on the keyboard. After responding, accuracy feedback was displayed for 500 ms, followed by a blank inter-trial interval

of 1,000 ms. Subjects completed 72 trials, and the dependent variable was the number of correctly identified target letters.

Selective visual arrays (Luck & Vogel, 1997; Shipstead et al., 2014; Figure 2). Subjects saw an array of blue and red rectangles differing in orientation. Prior to each trial, the subject was cued to attend to either the red or blue rectangles by a 300 ms flash of “RED” or “BLUE”. Next, the array was presented for 250 ms after a delay of 900 ms, and then the array was presented again with only the target rectangles (either red or blue). One of the target rectangles contained a white dot and changed orientation from the original array on 50% of the trials. The subject was asked whether or not the rectangle cued with the white dot had changed orientation from the initial presentation. Each array contained either 5 or 7 rectangles per color (10 and 14 total), and 40 trials were presented for each array size for a total of 80 trials. The dependent variable was a capacity score (k) calculated using the single probe correction (see Cowan et al., 2005; Shipstead et al., 2014). This calculation is $N * (\text{hits} + \text{correction rejections} - 1)$, where N is the set-size for that array. This calculation results in two separate k scores, one for set-size 5 and one for set-size 7, and the final dependent variable was the average k for these two set-sizes.

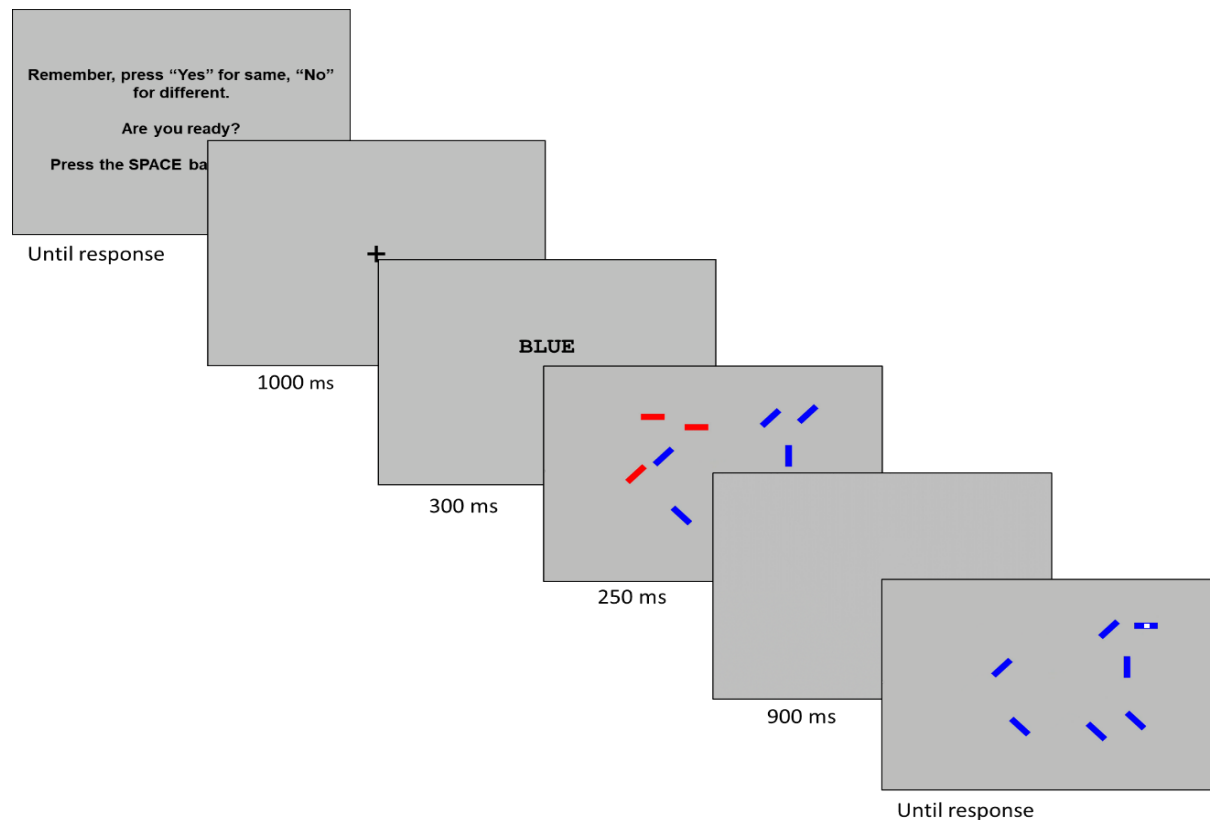


Figure 2 – Trial sequence for the selective visual array task. This example shows a trial of set-size 7. Here, the subject needed to indicated whether the blue rectangle in the upper-right of the array (indicated by a white dot in its center) had changed orientation from the previous display. In this trial, it did, and so the subject would respond by pressing the key which corresponded to the “Yes” response.

Psychomotor Vigilance Task. In this task, subjects were presented with a row of zeros in black font at the center of the screen (00000). After a variable wait time (equally distributed among 2, 4, 8, and 12 seconds) the zeros began to count up by 17 every 17 ms. The subjects were instructed to press the spacebar as quickly as possible once the numbers started counting up. After they pressed the spacebar, their reaction time was left on the screen for 1500ms to provide feedback. Eighty trials were administered. The dependent variable was the RTQ20, which is the average reaction time on each subject’s slowest 20% trials (Dinges & Powell, 1985; Unsworth & Robison, 2016).

Arrow Flanker (RT Flanker; Eriksen & Eriksen, 1974; Nieuwenhuis et al., 2006; Stoffels & van der Molen, 1988). Subjects were presented with a target arrow in the center of the screen pointing left or right along with two flanking arrows on both sides. The flanking arrows were either all pointing in the same direction as the central target (congruent trial; e.g., $\leftarrow \leftarrow \leftarrow \leftarrow \leftarrow$) or all in the opposite direction (incongruent trial; e.g., $\leftarrow \leftarrow \rightarrow \leftarrow \leftarrow$)⁶. The subject was asked to indicate direction the *central* arrow was pointing by pressing the “z” (left) or “/” (right) key. These keys had the words LEFT and RIGHT taped onto them to assist with response mapping. A total of 144 trials were administered; 96 congruent and 48 incongruent, with a randomized 400 – 700 ms ITI. The dependent variable was the flanker interference effect - the RT cost of the incongruent trials calculated by subtracting each subject’s mean RT on congruent trials from their mean RT on incongruent trials, excluding inaccurate trials.

Color Stroop (RT Stroop; Stroop, 1935). Subjects were shown the word “red” “green”, or “blue” in red, green, or blue font. The words were either congruent with the color (e.g., the word “red” in red print), or incongruent with the color (e.g., the word “red” in blue print). The subject’s task was to indicate which color the word was printed by pressing the “1”, “2”, or “3” key on the numpad. To assist with response mapping, the keys had a piece of paper of the corresponding color taped onto them. A total of 144 trials were administered; 96 congruent and 48 incongruent, with a randomized 400 – 700 ms ITI and a 5000 ms response deadline. The dependent variable was the Stroop interference effect – the RT cost of the incongruent trials calculated by subtracting each subject’s mean RT on congruent trials from their mean RT on incongruent trials, excluding inaccurate trials.

⁶ Note that the arrow flanker task often has three trial types: congruent, incongruent, and a neutral type in which dashes flank the central arrow. Similarly, in the Stroop task a neutral trial type is often present in which the word is not a color. Here we use only incongruent and congruent trials.

Sustained Attention-to-Cue Task (SACT; Figure 3). This is a novel task designed primarily as an accuracy-based version of the psychomotor vigilance task. In this task, subjects needed to sustain their attention on a visual circle cue presented at random locations on the screen and ultimately identify a target letter presented briefly at the center of the cue. The stimuli for the task were presented against a grey background. Each trial started with a central black fixation. On half of the trials, the fixation was presented for 2 seconds and for the other half the fixation was presented for 3 seconds. After the fixation, following a 300 ms tone, a large white circle cue was presented in a randomly determined location on either the left or right side of the screen. To orient the subject on the circle cue, the large circle began to immediately shrink in size until it reached a fixed size. Once the cue reached the fixed size, after a variable wait time (equally distributed amongst 2, 4, 8, and 12 seconds), a white asterisk meant to serve as a distractor appeared at the center of the screen. The asterisk blinked on and off in 100 ms intervals for a total duration of 300 ms (on for 100 ms, off for 100 ms, on for 100 ms). Then, a 3 x 3 array of letters was displayed at the center of the cue. The letters in the array consisted of B, D, P, and R. The central letter was the target letter and was presented in a dark grey font. The non-target letters were presented in black font with each letter occurring twice in the array and the target letter occurred three times. After 125 ms the central letter was masked with a “#” for 1000 ms. Only the central target letter was masked. After the mask, the response options were displayed in boxes horizontally across the upper half of the screen. The subject used the mouse to select whether the target was a B, D, P, or R. Feedback was given during the practice trials but not the experimental trials. Sixty-four trials were administered. Accuracy rate was the dependent variable.⁷

⁷ Although the sustained attention-to-cue task shares an asterisk distractor with the antisaccade, we believe that it

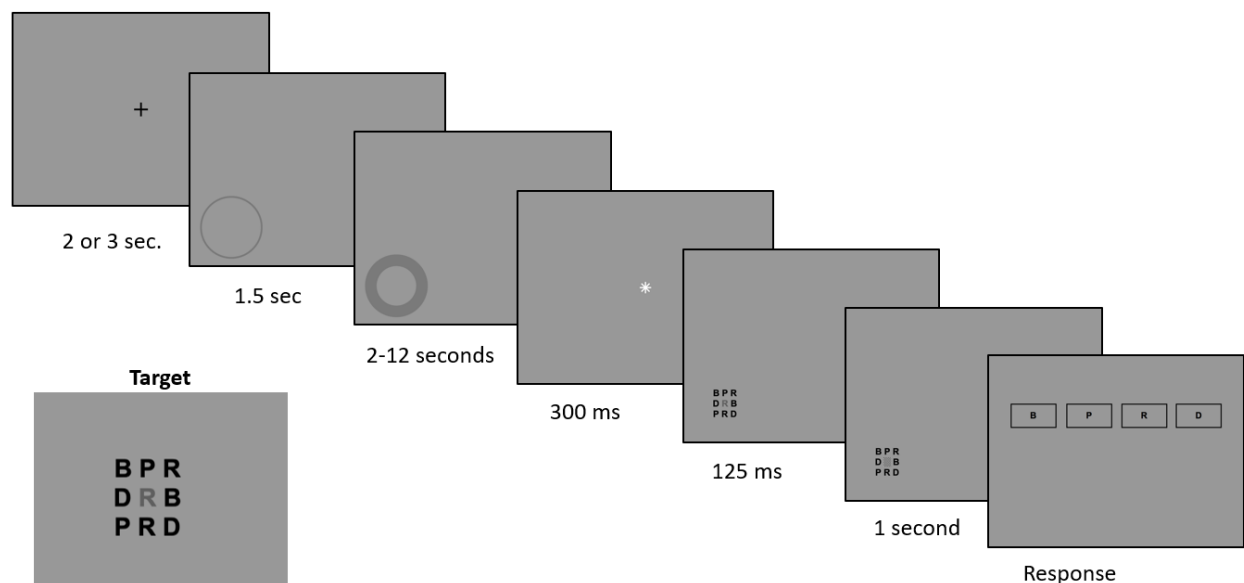


Figure 3 – Trial sequence for the sustained attention-to-cue task. Subjects saw a fixation for 2 or 3 seconds followed by a circle cue indicating the future location of the target letter array. This circle shrunk for 1.5 seconds and then remained for either 2, 4, 8, or 12 seconds. After this wait interval, a distracting asterisk then appeared outside of the circle for 300 ms. Then, the target 3 x 3 letter array (see enlarged view at bottom left corner) appeared for 125 ms and was then masked for 1000 ms. Subjects indicated which of four possible letters (B, D, P, or R) appeared in the center.

Adaptive Attention Control Tasks

Flanker Presentation Rate (Flanker PR). This task was a modified version of the arrow flanker and used an adaptive procedure to estimate the subject's threshold. Eighteen blocks of 18 trials each (total 324 trials) were administered. The duration that the stimuli appears on the screen (presentation rate) either decreased (shorter presentation time) if the subject was accurate

does not simply constitute a more complex version of antisaccade as suggested by an anonymous reviewer. First, as the name implies, this task involves a wait period in which the respondent presumably must sustain their attention at the cued location prior to stimulus onset. Second, a critical difference in the distractors in the antisaccade vs. this task is that the respondent has knowledge of the location of the target stimulus) *prior* to the presentation of the asterisk distractor in the sustained attention-to-cue task. In antisaccade, the respondent does not know where the target will be until the distractor is presented and must use the distractor to look to the opposite side of the screen. Therefore, in antisaccade the goal is to use the distractor as a cue to look in the opposite direction of said cue. In sustained attention-to-cue, it is to sustain attention over time on a particular pre-cued location while avoiding distraction.

on at least 15 trials within each block or increased (longer presentation time) if their accuracy rate was below that. The first block had a presentation time of 235 ms. For the first six blocks, the presentation time for the following block shortened in duration by 45 ms or increased in duration by 135 ms, again depending on whether or not the subject was accurate on at least 15 of the 18 trials. For subsequent blocks, the presentation time either quickened by 15 ms or slowed by 45 ms.⁸ Subjects could take as much time to respond as needed. Each block had 12 congruent and 6 incongruent trials in random order with a randomized 400 – 700 ms ISI. Congruent and incongruent trials were treated equally in determining if the presentation rate increased or decreased for the next block (i.e., subjects needed to be correct on 15 of 18 total trials independent of whether these trials were incongruent or congruent). The dependent variable was the presentation time after the final block of trials (i.e., what the presentation rate would have been on a hypothetical 19th block).

Flanker Deadline (Flanker DL). This task was a modified version of the arrow flanker and used an adaptive procedure to estimate the subject's threshold. Eighteen blocks of 18 trials each (total 324 trials) were administered. Each trial had a response deadline that limited how long the subject had to respond before they heard a loud beep and forfeited the opportunity to respond on that trial. This deadline either decreased (less time to respond) if the subject was accurate on at least 15 trials within each block or increased (more time to respond) if their accuracy rate was below that. The first block had a response deadline of 1050 ms. For the first six blocks, the response deadline either decreased by 90 ms or increased by 270 ms for the next block, again depending on whether or not the subject was accurate on at least 15 of the 18 trials.

⁸ If the presentation time were to be set below 10 ms., it was set to exactly 10 ms. instead.

For subsequent blocks, the response deadline either decreased by 30 ms or increased by 90 ms.⁹ The stimuli remained on the screen up until the response deadline. Each block had 12 congruent and 6 incongruent trials in random order with a randomized 400 – 700 ms ISI. The response deadline was the same for incongruent and congruent trials. Further, congruent and incongruent trials were treated equally in determining if the deadline increases or decreases for the next block (i.e., subjects needed to be correct and response before the response deadline on 15 of 18 total trials for the response deadline to decrease, independent of whether these trials were incongruent or congruent). The dependent variable was the response deadline after the final block of trials (i.e., what the deadline would have been on a hypothetical 19th block).

Stroop Deadline (Stroop DL). This task was a modified version of the Stroop and used an adaptive procedure to estimate the subject's threshold. Eighteen blocks of 18 trials each (total 324 trials) were administered. Each trial had a response deadline that limited how long the subject had to respond before they heard a loud beep and forfeited the opportunity to respond on that trial. This response deadline either decreased (less time to respond) if the subject was accurate on at least 15 trials within each block or increased (more time to respond) if their accuracy rate was below that. The first block had a response deadline of 1230 ms. For the first six blocks, the response deadline either decreased by 90 ms or increased by 270 ms for the next block, again depending on whether or not the subject was accurate on at least 15 of the 18 trials. For subsequent blocks, the response deadline either decreased by 30 ms or increased by 90 ms. The stimuli remained on the screen up until the response deadline.¹⁰ Each block had 12 congruent and 6 incongruent trials in random order with a randomized 400 – 700 ms ISI. The response deadline was the same for incongruent and congruent trials. Further, congruent and

⁹ If the response deadline were to be set below 150 ms, it was set to exactly 150 ms. instead.

¹⁰ If the response deadline were to be set below 150 ms, it was set to exactly 150 ms. instead.

incongruent trials were treated equally in determining whether or not the deadline increased or decreased for the next block (i.e., subjects needed to be correct and response before the response deadline on 15 of 18 total trials for the response deadline to decrease, independent of whether these trials were incongruent or congruent). The dependent variable was the response deadline after the final block (i.e., what the deadline would have been on a hypothetical 19th block).

Adaptive Visual Cue Task (AVC; Figure 4). This was a new task. A valid cue was presented where a target line was to be located within a larger search array that was presented for a brief duration. Subjects needed to identify whether the target line was of vertical, horizontal, or diagonal orientation. The search array was composed of 124 distractor lines of vertical, horizontal, or diagonal orientation. The orientation and location of each distractor line was randomly determined for each trial. The duration of the cue adaptively changed on a trial-by-trial basis. The trial sequence was as follows. Following a 300 ms tone, a white asterisk appeared on either the far left or far right side of the screen. The asterisk blinked on and off in 100 ms intervals for a total duration of 300 ms (on for 100 ms, off for 100 ms, on for 100 ms). A blue circle cue was then displayed on the opposite side of the screen for a variable duration (determined by the adaptive procedure). Then, the search array of the target and distractor lines was presented for 150 ms, after which each stimuli on the display was masked for 300 ms. The subject then used the mouse to select whether the target line was of vertical, horizontal, or diagonal orientation. Feedback was given during the practice trials but not the experimental trials.

We used weighted-up-down adaptive procedure with 64 trials to estimate a difference threshold to converge on a certain level of performance (Kaernbach, 1991). If subjects made a correct response, the duration of the cue decreased on the next trial (less time to orient to the

visual cue). If the subject made an error, the duration of the cue increased on the next trial (more time to orient to the visual cue). The ratio to decrease/increase the duration was 1:2. This weighted-up-down method with a step ratio of 1:2 converges around a threshold value of 66% (Kaernbach, 1991). To estimate this threshold value, we employed a method of averaging the difference value for the last four reversals (Hairston & Maldjian, 2009). A reversal is each instance of which the accuracy on the current trial was different from the accuracy on the previous trial. The critical dependent measure was the average difference value of the last four reversals.

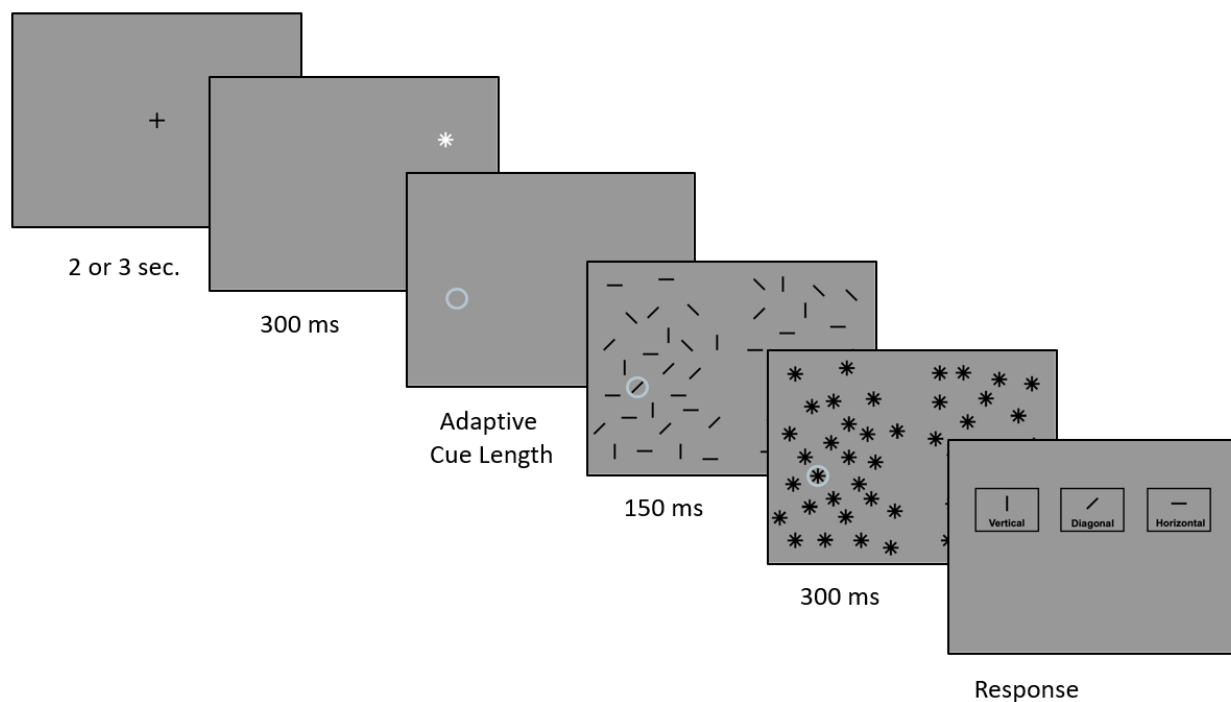


Figure 4 – Trial sequence for the adaptive visual cue task. A central fixation appears for 2 or 3 seconds, followed by the appearance of an asterisk at a random location on either the left or right side of the screen for 300ms. Following the asterisk, a cue appears on the opposite side of the screen for a variable amount of time. Then the visual search array appears, with a target line at the center of the cue in one of three orientations, vertical, diagonal, or horizontal. After 150ms the visual search array is masked for 300ms. The subject must select which orientation the target had. The length of the cue is determined on a trial-by-trial basis using an adaptive up-down procedure. After a trial with a correct response, the length of the cue is shorter on the following trial. After a trial with an incorrect response, the length of the cue is longer on the following trial.

Data Preparation and Analysis

Scoring Criteria

Data processing steps were implemented at multiple stages to prepare the data for statistical analysis. To outline, data were first trimmed at the task level data depending on task-specific characteristics and we removed subjects' scores on a task if they showed poor performance indicating they did not understand the instructions or were not truly performing the task (the criteria for each task is described below). Next, for each task separately we removed outlier scores (± 3.5 standard deviations). These two steps can lead to a subject having missing data on some tasks and not others. Missing data also occurred for a variety of other reasons including lost data file, experimenter error, or if the task crashed during administration, which accounted for less than 1% of total scores. Finally, for the working memory capacity and fluid intelligence tasks, if a subject had missing data on more than one task for either construct they were completely removed from any further analyses. This criterion was not used for the attention control factor because a more exploratory approach was taken to determine which tasks are the best attention control measures. After removing scores based on these criteria, a total of 7 subjects were completely removed from further data analysis for a total sample size of 396. Out of the 396, no single task had more than 5% scores missing.

For the working memory tasks, if a subject's mean accuracy on the processing portion of the task was 3.5 standard deviations below the mean, then their score on the task was removed.

For the standard Stroop and flanker tasks, we removed trials that had extreme reaction time values. The purpose of this was to remove trials with too short of reaction times to accurately reflect task processing. If reaction times were shorter than 200 ms, the trial was removed. If the subjects mean accuracy on congruent or on incongruent trials was 3.5 standard deviations below the mean across all subjects, then their score on the task was removed.

For the psychomotor vigilance task, we removed trials that had extreme reaction time values. Again, the purpose of this was to remove trials with too short or too long of reaction times to accurately reflect task processing. If reaction times were shorter than 200 ms or longer than 10 seconds, the trial was removed.

For the visual arrays, Flanker DL, Flanker PR, Stroop DL, and Adaptive visual cue tasks, if a subject's mean accuracy on the task was 3.5 standard deviations below the mean their score on that task was removed.

Data Analysis

We used five criteria to evaluate whether each new and modified attention control tasks improve the measurement of attention control; (1) reliability of each task, (2) inter-correlations with other attention control tasks, (3) latent factor loadings, and (4) external validity to working memory capacity and fluid intelligence. The fifth criterion, mediation of the working memory capacity/fluid intelligence relationship, was not used to assess tasks individually but rather whether the new and modified tasks were improvements in aggregate.

For reliability, we assessed internal consistency (for tasks that were appropriate in doing so) using split-half reliability (even/odd trials) corrected using the Spearman-Brown prophecy

formula. Internal consistency was not calculated for adaptive and threshold-based measures, as this did not seem appropriate given the nature of these tasks. We assessed test-retest reliability in all attention tasks by bringing back a subset of subjects for an additional two-hour session. Attention measures were administered in the same relative order to one another in the retest session as the initial testing sessions. All attention measures could not fit within one two-hour session, so we created two separate conditions for test-retest reliability. Antisaccade, visual arrays, and sustained attention-to-cue were included in both conditions whereas the other tasks were only included in one, resulting in a large discrepancy in terms of test-retest sample sizes across the tasks. In hindsight, it was perhaps a mistake on our part not to include all new and modified measures in both test-retest conditions as it limited our ability to assess reliability in the adaptive tasks. That is, because internal consistency could not be calculated for the adaptive tasks (namely the modified Stroop and flanker tasks), the only estimate of reliability we have for these tasks is based on test-retest reliability for 58 - 66 subjects with, on average, 194 days between initial administration of the tasks and retest. Readers are therefore urged to exercise caution in interpreting the reported test-retest reliabilities, particularly for the tasks with relatively few subjects.

The factor loadings are based on an exploratory structural equation modelling approach. The rationale for this was that in addition to including a model with all 10 attention tasks loaded onto a single attention control factor, we wanted to evaluate what combination of three tasks forms the strongest attention control factor. The use of at least three tasks on a latent factor is standard as it helps avoid issues with under-defined models and it is not always practical for researchers to include more than three indicators for a latent variable given time and resource constraints of behavioral research. For the 10 tasks, there are 120 possible combinations of three

tasks. Therefore, we ran 120 structural equation models with an attention control factor predicting working memory capacity and fluid intelligence.

For external validity, we used structural equation modeling to test whether attention control fully mediated the relationship between working memory capacity and fluid intelligence. We present several models in comparison to our typical attention control factor with the antisaccade, flanker, and Stroop (reaction time difference score versions).

For all structural equation models, solid paths represent statistically significant paths ($\alpha = .05$) and dotted lines represent statistically non-significant paths. The chi square values, degrees of freedom, and chi square significance are reported. The chi-square assesses overall fit and discrepancy between the sample and generalized population wide fitted covariance matrices (i.e., how far apart is the covariance matrix implied by the model to the observed covariance matrix). Although a non-significant chi-square value is preferred, indicating that the model is not statistically different from a general population model, it is very sensitive to sample size and degrees of freedom, and the chi-square test is based on assumptions which are rarely met in practice (see Schermelleh-Engel, Moosbrugger, & Müller, 2003). As such, the chi-square value alone is not sufficient to accept or reject a model and is often instead used as one source of evidence for model fit (Anderson & Gerbing, 1988; Schermelleh-Engel et al., 2003). Models must therefore be considered in holistic terms based on multiple fit indices, namely the confirmatory fit indices (CFI) and the Root Mean Square Error of Approximation (RMSEA). The CFI compares model fit to a null model. Models with a CFI $> .90$ are considered an acceptable fit, and with a CFI of .95 or higher considered to be a good fit (Hu & Bentler, 1999). The RMSEA is a parsimony adjusted fit index. Models with an RMSEA $< .08$ are considered to be an acceptable fit, with an RMSEA of .05 or lower considered to be a good fit (Browne &

Cudeck, 1993; Kenny, 2015) although Hu and Bentler recommended .06 as the cutoff instead. Any models that are not considered as an acceptable or good fit are thereby considered as a poor model fit, suggesting that the model does not sufficiently recreate the observed covariance matrix.

A nuance to assessing model fit in structural equation modeling is the balance between model parsimony/generalizability and a well-fitting model. Most software used to perform structural equation modeling reports some combination of the Wald test, Lagrange multiplier test, and likelihood ratio test which all assess how specifying the model differently would alter model fit and list changes in descending order of the impact of each change. Using these tests to adjust parameters and constraints to achieve a better fitting model is controversial and can lead to a practice known as overfitting. An over-fit model will have excellent fit indices but also a large number of post-hoc parameters and constraints that are specific to the particular data set and will likely not generalize to other data sets or populations. In addition, it can often be difficult to theoretically justify certain post-hoc modifications, such as why two seemingly poorly related indicators should have correlated error terms for any reason other than it improved model fit. It is therefore our practice to report the most parsimonious models (e.g., ones without a number of cross-loaded indicators and correlated error terms) and to engage in post-hoc re-specification of the model only when absolutely necessary or justified.

Finally, in some analyses we report whether a path value from one model is significantly different than a path value in another. This is done using the model comparisons approach (Loehlin, 1987) in which nested models are tested against one another. If the model with fewer constraints (more degrees of freedom) is not significantly different from the nested (more constrained) model in terms of fit, then the model with more degrees of freedom is preferred due

to parsimony. However, if the nested (more constrained) model is statistically a better fit to the data than the less constrained, the nested model is preferred as it does a better job of recovering the covariance matrix of the original data.

All analyses were conducted in R statistical software (R Core Team, 2018). The R package *lavaan* (Rosseel, 2012) was used for all structural equation analyses, treatment of missing values was set to full-information maximum likelihood.

Results

Descriptive statistics for all relevant tasks, including time of administration for the attention measures, are provided in Table 1.

Table 1

Descriptive Statistics of dependent variables from each task

Task	Mean	SD	Skew	Kurtosis	Admin Time (minutes)
Raven	10.26	3.22	-.41	-.27	-
LetterSets	17.09	4.34	-.32	-.31	-
NumSeries	9.81	3.09	-.40	-.48	-
Operation Span	55.76	15.58	-.73	.08	-
Symmetry Span	27.90	10.35	-.06	-.40	-
Rotation Span	24.49	9.30	-.07	-.29	-
Visual Arrays	1.86	1.17	.19	-.59	13.0
Antisaccade	.79	.15	-.90	.03	9.8
Psychomotor Vigilance	889.44	754.68	2.77	7.72	26.8
Sustained Attention-to-Cue	.70	.19	-.76	-.06	24.8
Adaptive Visual Cue	58.83	42.92	4.91	38.39	12.0
Stroop DL	1013.01	345.33	1.97	5.56	10.1
Flanker DL	674.61	212.90	2.33	6.97	10.5

Flanker PR	101.35	93.4	1.73	2.80	14.5
RT Flanker Effect	80.69	43.18	1.08	1.98	9.1
RT Stroop Effect	131.62	85.53	.73	.83	5.7

Note. Stroop DL = modified Stroop with adaptive response deadline; Flanker DL = modified flanker with adaptive response deadline; Flanker PR = modified flanker with adaptive presentation rate. Admin. time = 95th percentile in terms of time it took subjects to complete the task. That is, 95% of subjects finished at or before that time.

Criterion 1 - Reliability

Internal consistency and test-retest reliabilities for the attention control measures are presented in Table 2. The psychomotor vigilance, sustained attention-to-cue, and antisaccade tasks were highly reliable in terms of internal consistency (.92 - .95). The visual arrays had lower internal consistency (.75) but ranked higher than the reaction time difference score versions of the flanker (.74) and Stroop (.69). In terms of test-retest reliability, the antisaccade, visual arrays, response deadline Stroop, and sustained attention-to-cue ranked highest (.63 - .73). While the psychomotor vigilance task showed high internal consistency, the test-retest reliability was .55. The modified flanker with adaptive response deadline had a poor test-retest reliability (.54) though this was higher than the reaction time Stroop (.46) and reaction time flanker (.23). The adaptive visual cue had a very poor test-retest reliability (.39) as did the modified flanker with adaptive presentation rate (.32).

Reliability and stability

The second administration of these tasks occurred, on average, over six months after the initial administration ($M = 194$ days; $SD = 131$; min = 6; max = 419). Because of this and the large variance in number of days elapsed between test and retest, we were concerned that number of days elapsed could have affected test-retest reliability. We conducted a moderation analysis with number of days as the interaction term and found no statistically significant moderation of

number of days elapsed on test-retest reliability for any of the measures (accounting for only around 1% of the variance), indicating that number of days did not interact with the stability of the test-retest estimate. This was surprising given that the test-retest reliabilities were overall very low for most tasks, but results may have been different if the retest session had occurred within a week or so of initial administration and if the sample size was larger.

Table 2

Internal consistency and test-retest reliabilities for attention tasks

Task	I.C. Initial (n)	I.C. Retest (n)	Test-Retest	Retest w/o Outliers
PVT	.95 (388)	.92 (63)	.55	.55
SACT	.93 (383)	.95 (126)	.52	.63
Antisaccade	.92 (390)	.92 (126)	.71	.73
Visual Arrays	.75 (397)	.71 (125)	.67	.69
RT Flanker	.74 (385)	.70 (71)	.23	.23
RT Stroop	.69 (386)	.62 (67)	.46	.46
Stroop DL	-- (390)	-- (66)	.55	.67
Flanker PR	-- (386)	-- (58)	.32	.32
Flanker DL	-- (382)	-- (66)	.31	.54
AVC	-- (386)	-- (65)	.31	.39

Note. PVT = psychomotor vigilance task; SACT = sustained attention-to-cue; Stroop DL = modified Stroop with adaptive deadline; Flanker PR = modified flanker with adaptive presentation rate; Flanker DL = modified flanker with adaptive response deadline; AVC = adaptive visual cue. Internal consistency (I.C.) was calculated using an even-odd split procedure and corrected using the Spearman-Brown prophecy formula. I.C. Initial is the internal consistency for the first administration of the task. I.C. Retest is the internal consistency for the second administration (retest). Test-retest is the correlation between performance on the initial and retest administrations. Retest without Outliers is this correlation after removing outliers for each task. Outliers are defined as subjects with a Z-score difference greater than 3 between their rank order in the first and second administration. Average days elapsed between administration 1 and administration 2 was 194 ($SD = 131$). We do not present reliability estimates for working memory capacity or fluid intelligence tasks as the psychometric properties of these tasks have been established (e.g., Conway et al., 2005), but these measures all had internal consistencies in the .80s except the rotation span partial score (.77).

Criterion 2 – Inter-correlations among tasks

The full correlation matrix for the attention control measures is presented in Appendix A (Table A) and Table 3 shows a summarized version of the average inter-correlation among these

measures. The reaction time flanker and Stroop tasks (based on difference scores) had the lowest correlations with other attention measures, which is consistent with previous research. On the other hand, the antisaccade and sustained attention-to-cue tasks had the strongest average inter-correlations of $r = .35$ and $.32$, respectively, with all other tasks in the $r = .18 - .27$ range.

Because we consider the antisaccade to be a hallmark measure of attention control, we also present in Table 3 the correlation of each task to the antisaccade. In terms of linear association with the antisaccade, all correlations were statistically significant ($p < .05$). Performance on the selection visual arrays and sustained attention-to-cue tasks correlated quite strongly to antisaccade ($r = .45$ and $.40$, respectively), though not as strong as the inter-correlations among tasks of working memory capacity or fluid intelligence which are generally in the $.50$ s. The psychomotor vigilance task, adaptive visual cue task, modified flanker with an adaptive response deadline, and modified Stroop with an adaptive response deadline also had relatively strong correlations to the antisaccade ($r = .31 - .35$). The modified flanker with an adaptive presentation rate had a moderate correlation to antisaccade ($r = .25$), and the reaction time difference score flanker and Stroop had the weakest correlations to antisaccade ($r = .16$ and $.19$, respectively).

Table 3

Inter-correlations among the attention control measures

Task	Average Correlation	Correlation to Antisaccade
Antisaccade	.35	--
SACT	.32	.40
Visual Arrays	.27	.45
PVT	.26	.35
Flanker DL	.25	.34
Flanker PR	.25	.25
Stroop DL	.24	.31
AVC	.18	.33
RT Flanker Effect	.15	.16
RT Stroop Effect	.11	.19

Note. PVT = psychomotor vigilance task; SACT = sustained attention-to-cue; RT Stroop and RT flanker are reaction time difference scores (Stroop and flanker effect, respectively); Stroop DL = modified Stroop with adaptive deadline; Flanker PR = modified flanker with adaptive presentation rate; Flanker DL = modified flanker with adaptive response deadline; AVC = adaptive visual cue. Correlations involving one task in which lower scores indicate better performance (e.g., reaction time) were multiplied by -1 such that a positive correlation indicates that individuals who performed better on one task also performed better on the other. All tasks statistically significantly correlated to antisaccade. $N = 382 - 390$.

*Criterion 3 –Latent coherence**Exploratory factor analysis*

We ran an exploratory factor analysis using principal axis factoring with a varimax rotation and not specifying a set number of factors (Table 4). Visual inspection of the scree plot suggested that two factors were sufficient; one factor for fluid intelligence and working memory measures, and another for the attention measures. Five factors had eigenvalues above 1. While using the eigenvalue above 1 criterion is likely not the best method of choosing numbers of factors (e.g., Osborne, Costello, & Kellow, 2008), in this case it was more informative than forcing 2 or even 3 factors. We ran additional models either specifying a certain number of factors and/or with oblique, rather than orthogonal, rotations. The models converged upon the following conclusions; 1) The working memory capacity tasks and fluid intelligence tasks formed their own separate factors which, in the orthogonal models, accounted for the most variance, 2) The antisaccade, sustained attention-to-cue, psychomotor vigilance task, adaptive visual cue task, visual arrays loaded together on a separate factor that accounted for the third most variance among the factors, 3) All three flanker measures loaded onto their own factor, driven by the adaptive response deadline flanker task (though any flanker task could be removed from the analysis and the other two still loaded together and separately from the other tasks), 4) Performance on the reaction time Stroop task loaded onto a separate factor with the modified Stroop with response deadline (which also equally loaded onto the attention control factor), 5) Performance on the visual arrays loaded more strongly with working memory capacity and fluid intelligence than other attention tasks did, generally with a similar magnitude as the visual arrays' loading with the other attention measures, and, 6) If we forced fewer factors, the working memory capacity and fluid intelligence measures would form a single *g* factor before the flanker

measures would load with the rest of the attention measures. As a check to the validity of these results, we also added sensory discrimination measures to the exploratory model (data not reported here), which resulted in an additional factor comprised of just these three discrimination tasks having an eigenvalue above 1.

Table 4

Exploratory factor analysis – rotated factor loadings

Task	Factor 1	Factor 2	Factor 3	Factor 4	Factor 5
Raven	.26	.55	.13	-.12	-.15
LetterSets	.20	.67	.05	-.09	-.09
NumSeries	.18	.73	.07	-.05	-.01
Operation Span	.53	.30	-.04	-.06	.06
Symmetry Span	.86	.19	.07	-.16	-.08
Rotation Span	.62	.31	.19	-.17	-.06
Visual Arrays	.30	.27	.37	-.03	-.20
Antisaccade	.20	.25	.50	-.15	-.18
Psychomotor Vigilance	.03	-.02	.60	.12	-.08
Sustained Attention-to-Cue	.12	-.02	.55	-.13	-.01
Adaptive Visual Cue	-.03	.04	.38	.00	-.05
Stroop DL	.14	.12	.23	.16	.24
Flanker DL	-.05	.16	.09	.90	-.01
Flanker PR	.19	-.01	.25	.40	.08
RT Flanker Effect	-.06	-.05	.05	.36	.09
RT Stroop Effect	.01	.12	.02	.14	.86

Note. Stroop DL = modified Stroop with adaptive response deadline; Flanker DL = modified flanker with adaptive response deadline; Flanker PR = modified flanker with adaptive presentation rate. Extraction done via principal axis factoring with varimax rotation. The strongest loading for each task is in boldface. Loadings at or above .25 are shown in italics if the loading is not the strongest for that particular task. For ease of interpretation, some loadings were multiplied by -1 such that positive loadings reflect better performance for that task was positively related to the factor.

We tested the attention control measures in all possible combinations of a three-indicator factor (120 unique models, each task appeared in 36 of them) and when predicting both fluid intelligence and working memory capacity. Table 5 shows a summary of the average loading for each task across these 120 models. The antisaccade and visual arrays had very strong average loadings (.75 and .69, respectively), whereas the adaptive visual cue task, reaction time flanker, and reaction time Stroop had, on average, very low loadings (.28 - .31) which are below acceptable levels. The other tasks are in the middle, ranking from .40 to .59.

Table 5

Average factor loadings for each attention task

Task	Average Loading (<i>SD</i>)
Antisaccade	.76 (.08)
Visual Arrays	.69 (.06)
Flanker DL	.59 (.12)
Sustained Attention-to-Cue	.51 (.11)
Flanker PR	.48 (.08)
Stroop DL	.46 (.08)
Psychomotor Vigilance	.40 (.09)
Adaptive Visual Cue	.31 (.07)
RT Flanker	.31 (.08)
RT Stroop	.28 (.07)

Note. SACT = sustained attention-to-cue; Stroop DL = modified Stroop with adaptive deadline; Flanker PR = modified flanker with adaptive presentation rate; Flanker DL = modified flanker with adaptive response deadline. The average loading was calculated by taking the average absolute value of each task's loading across all 120 possible tri-task factors in which each task appeared 36 times.

As it is not practical to show all 120 tri-indicator attention control factors, we instead present a model with all attention tasks loaded onto a single factor to summarize the latent results (Figure 5). The rank-ordering of the factor loadings for this model is generally consistent with Table 5, and the loadings are quite similar to the average loading for each task across all 120 combinations of tri-indicator attention control factors (also Table 5). Note that this model is only for illustrative purposes to show the relative loadings of the attention tasks on the attention control construct. It is not a particularly good model to assess attention control because a number of poor indicators are present and it is likely that model fit would be improved by loading a subset of the attention tasks onto subfactors of attention. As a result, model fit is relatively poor (e.g., CFI = .90) and the mediation of attention control on the working memory capacity/fluid intelligence relationship is worse than when fewer indicators are used instead (see Figure 6).

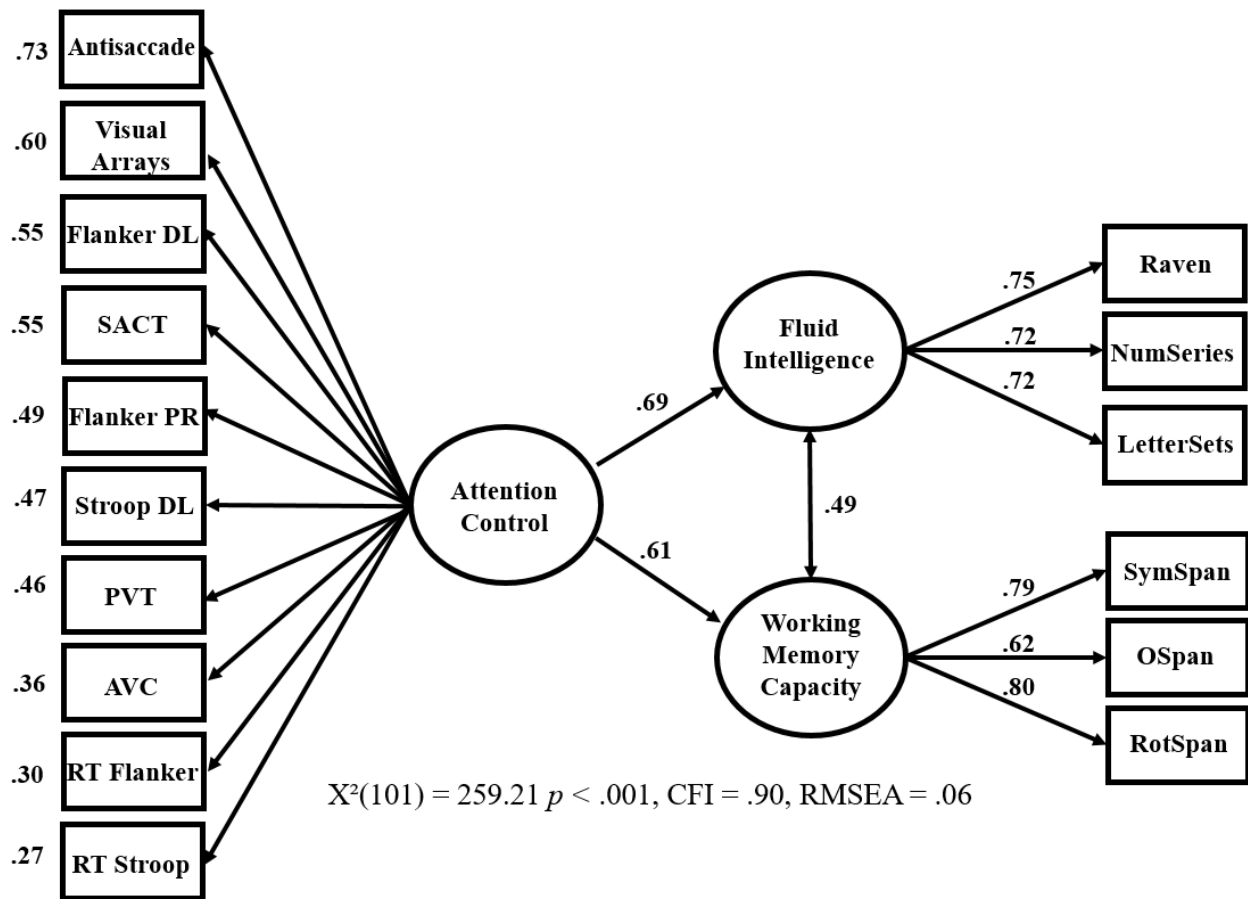


Figure 5 - Structural equation model with all attention tasks loaded onto a single factor. PVT = psychomotor vigilance task; SACT = sustained attention-to-cue; Stroop DL = modified Stroop with adaptive deadline; Flanker PR = modified flanker with adaptive presentation rate; Flanker DL = modified flanker with adaptive response deadline; AVC = adaptive visual cue; NumSeries = number series; SymSpan = symmetry span; OSpan = operation span; RotSpan = rotation span. Factor loadings for the attention control measures are on the left. Loadings and paths involving tasks in which lower scores indicate better performance (e.g., reaction time) were multiplied by -1. The correlation between working memory capacity and fluid intelligence is the correlation between their disturbance terms. $N = 396$.

Criterion 4 – Relationship to working memory capacity and fluid intelligence

The final criterion for individual task improvement is attention control's relationship to fluid intelligence and working memory capacity. According to our theory of working memory capacity as executive attention, the strong link between working memory capacity and fluid intelligence is primarily due to individual differences in attention control (Engle, 2002; Engle &

Kane, 2004). According to this theory, attention control underlies all higher-order and goal-directed behavior, and hence it should strongly associate with, potentially even fully mediate, the relationship between any two executive functions such as working memory capacity and fluid intelligence. However, potentially due to aforementioned problems with assessing individual differences in attention control, a full mediation of attention control on the working memory capacity-fluid intelligence relationship has not been observed. It has been argued that abilities such as secondary memory and memory updating are required, along with attention control, to fully explain the relationship between working memory capacity and fluid intelligence, and yet even when these other constructs are tested there is still variance unaccounted for (e.g., Unsworth & Spillers, 2010). We would argue that attention control alone should be sufficient to explain the variance in the working memory capacity and fluid intelligence relationship, provided that attention control is measured properly.

Table 6 shows the first-order correlations between all the attention control tasks with composite Z-scores of working memory capacity and fluid intelligence. Note that the working memory capacity and fluid intelligence composites correlated at $r = .55$ and their latent scores correlated at $r = .67$ (45% of their reliable variance was shared at the latent level). Table 6 shows that the visual arrays and antisaccade tasks quite strongly correlate with working memory capacity and fluid intelligence ($r = .41 - .46$), and that the modified flanker with an adaptive response deadline has the next strongest correlation ($r = .29$ with working memory capacity, $r = .33$ with fluid intelligence). The only attention measure that did not correlate significantly with working memory capacity was the reaction time Stroop ($r = .10$, $p = .054$), although this correlation was not statistically different than the psychomotor vigilance's ($r = .12$) or adaptive visual cue task's ($r = .10$) correlation to working memory capacity. All measures significantly

correlated with fluid intelligence and had correlations at or above $r = .18$. The modified flanker with an adaptive presentation rate, modified Stroop with adaptive response deadline, sustained attention-to-cue task, and reaction time flanker correlated similarly with working memory capacity ($r = .18 - .23$) and fluid intelligence ($r = .16 - .23$) composite scores.

Table 6

Correlations between attention measures and working memory capacity/fluid intelligence

Task	Correlation to WMC	Correlation to Gf
Visual Arrays	.43*	.46*
Antisaccade	.41*	.46*
Flanker DL	.29*	.33*
Flanker PR	.23*	.22*
Stroop DL	.21*	.26*
SACT	.21*	.23*
RT Flanker Effect	.18*	.16*
PVT	.12*	.20*
AVC	.09	.18*
RT Stroop Effect	.09	.21*

Note. WMC = working memory capacity; Gf = fluid intelligence; PVT = psychomotor vigilance task; SACT = sustained attention-to-cue; Stroop DL = modified Stroop with adaptive deadline; Flanker PR = modified flanker with adaptive presentation rate; Flanker DL = modified flanker with adaptive response deadline; AVC = adaptive visual cue. Correlations involving one task in which lower scores indicate better performance (e.g., reaction time) were multiplied by -1 such that a positive correlation means individuals who performed better on one task also performed better on the other. WMC and Gf are Z-score composites. $N = 378 - 397$.

* $p < .05$.

Criterion 5 – Mediation of the working memory capacity/fluid intelligence relationship

Another way to test the relationship between attention control and working memory capacity/fluid intelligence is through latent mediation models with attention control mediating the working memory capacity/fluid intelligence relationship. One issue with this approach is that latent analyses require multiple indicators per factor, and so comparing one task to another is less straightforward. Nonetheless, these mediation models can provide a better understanding of how the new and modified attention tasks performed in aggregate.

We used results reported thus far to inform our decision of which models to test, and these four models are presented in Figure 6. The first model we tested (Figure 6a) was the baseline model consisting of three attention measures used in our previous papers; the antisaccade, RT Stroop, and RT flanker. Of note in this model is that the RT flanker and Stroop tasks had poor loadings (.28 and .30, respectively) whereas the antisaccade had a strong loading (.68). This indicates that the antisaccade dominated the factor (a common finding as we discussed previously and reported in Rey-Mermet et al., 2018) such that the attention control factor is mostly composed of variance from this single task. Also, of note is that the attention control factor did not fully mediate the relationship between working memory capacity and fluid intelligence, as the path between the two (.33) was statistically significant even after accounting for attention control.

The second model (Figure 6b) has the best performing attention tasks per the previous criteria (antisaccade, sustained attention-to-cue, and adaptive response deadline flanker). Here, the antisaccade and visual arrays had statistically equal loadings (.67 and .64, respectively), and the flanker deadline task loaded at .46. The attention factor had a strong relationship to working memory capacity (.75) and fluid intelligence (.65). Importantly, attention control did fully

mediate the working memory capacity/fluid intelligence relationship in this model (the path from working memory capacity to fluid intelligence is .21 and non-significant). However, setting this path to .33 (the value from the first model) and using the model comparisons approach to test whether this .21 was statistically significantly lower than .33 indicated that this difference was *not* statistically significant. As such, attention control in this model did not statistically significantly mediate the working memory capacity/fluid intelligence relationship more than when the antisaccade, reaction time Stroop, and reaction time flanker were used.

The antisaccade was an indicator for attention control in the models from Figure 6a and Figure 6b. While we argue that the antisaccade is a great measure of attention control because of its simplicity and high validity, it is not ideal to rely on one task for understanding attention control more broadly. Also, it is possible that the antisaccade is anchoring the results of these two models. So, in the third model (Figure 6c), the antisaccade was excluded and instead we tested the other best-performing tasks (visual arrays, modified flanker with adaptive response deadline, and sustained attention-to-cue). In this model, factor loadings to the attention factor are worse than the second model (.64 for visual arrays, .44 for modified flanker with adaptive response deadline, and .38 for sustained attention-to-cue), however these are still improved over the reaction time Stroop and flanker tasks from the first model. Further, these three tasks fully mediated the working memory capacity/fluid intelligence relationship (the path was .16 and non-significant). Using the model comparisons approach, this was statistically significantly lower than the .33 from the first model. This indicates that the visual arrays, flanker deadline, and sustained attention-to-cue did statistically significantly improve attention control's mediation of working memory capacity/fluid intelligence relationship over the more traditional attention tasks used in Figure 6a (antisaccade, reaction time Stroop, reaction time flanker).

In the fourth model (Figure 6d) we tested attention control's mediation of the working memory capacity/fluid intelligence with attention control comprised of all the reaction time measures – the psychomotor vigilance task and standard reaction time difference score Stroop and flanker. This model was conducted to alleviate potential concerns that part of the reason these tasks performed relatively poorly compared to most of the accuracy-based ones is because the number of accuracy-based measures far outweighed the number of reaction time ones (7:3). But, these three reaction time measures cohered quite poorly, to the extent that, without imputation and depending on how data filtering was performed, there were issues with non-statistically significant factor loadings and model convergence. However, we were able to get the model to converge. Still, the loadings were low (.29 - .42), and the paths from working memory capacity to attention control and from attention control to fluid intelligence were much weaker than in the other models (.48 and .43, respectively). This model was the weakest in terms of mediating the working memory capacity/fluid intelligence relationship, as the path between these two was .49 and statistically significant. Using model comparison, this .49 mediation value was statistically worse than in any other mediation model shown in Figure 6, indicating that the reaction time attention tasks had the weakest statistical mediation of the working memory capacity/fluid intelligence relationship.

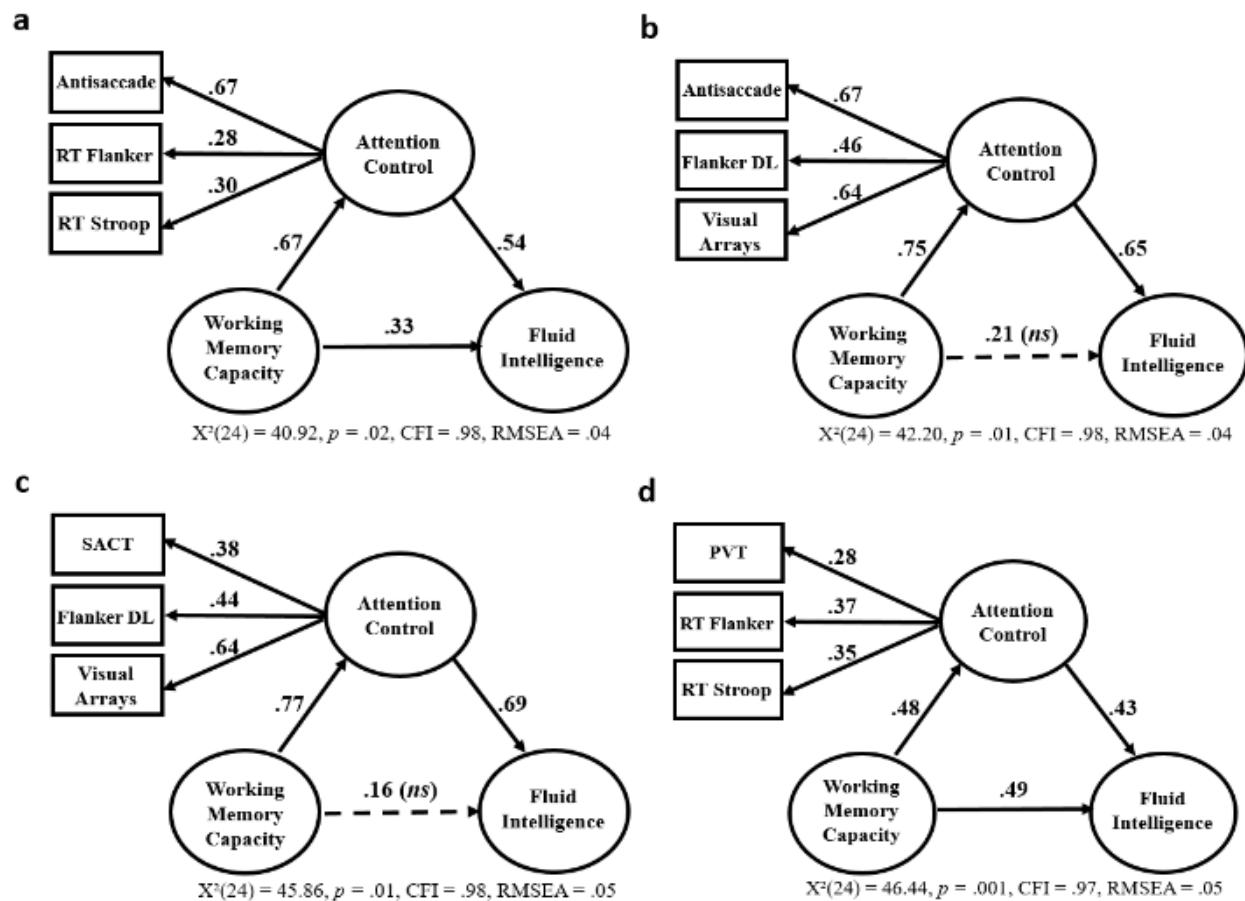


Figure 6 - Structural equation models with attention control mediating the working memory capacity/fluid intelligence relationship. Different models are compared in which attention control is comprised of a) traditional indicators - antisaccade, RT flanker, and RT Stroop; b) the best three indicators based on reliabilities, intercorrelations, and relationship to working memory capacity and fluid intelligence - antisaccade, Flanker DL, and visual arrays; c) the best three indicators excluding antisaccade - SACT, Flanker DL, and visual arrays; and d) indicators with reaction time and reaction time difference scores as the dependent variable – PVT, RT flanker, and RT Stroop. Flanker DL = Modified flanker with adaptive response deadline. SACT = sustained attention-to-cue task. PVT = psychomotor vigilance task. Working memory capacity was comprised of symmetry span, operation span, and rotation span. Fluid intelligence was comprised of Raven's advanced, number series, and letter sets. Non-significant paths are shown with a dashed line. $N = 396$.

Additional analyses

We also performed three sets of post-hoc analyses designed to address unresolved concerns of our own or raised during the review process. The first was informed by the exploratory factor analysis results in which we tested whether the three flanker tasks formed a

coherent latent factor separate from the rest of the attention measures. The second was a test of the extent to which processing speed or general processing could account for the improvement of the new and modified attention tasks. The third set of analyses was conducted to explore whether using pure reaction times on congruent and incongruent trials of the Stroop and flanker as the dependent variable would be a viable alternative to the Stroop and flanker tasks with adaptive response deadlines or presentation rate.

Exploratory factor analysis indicated that the flanker tasks could load onto their own factor separate from the other attention measures. We tested a structural equation model with the flanker tasks loaded onto a separate factor from the other attention measures, and both of these factors predicting working memory capacity and fluid intelligence (Figure 7). The model fit the data well and indeed the flanker tasks formed a coherent separable factor. However, this factor strongly correlated with the other attention control factor (.64) and did not statistically significantly predict working memory capacity or fluid intelligence above and beyond other attention measures, meaning that the flanker tasks did not capture unique variance in working memory capacity or fluid intelligence above the other attention tasks. This indicates that the flanker factor reflects flanker-specific variance that is separable from other attention measures but is not important for explaining variance in the working memory capacity/fluid intelligence relationship.

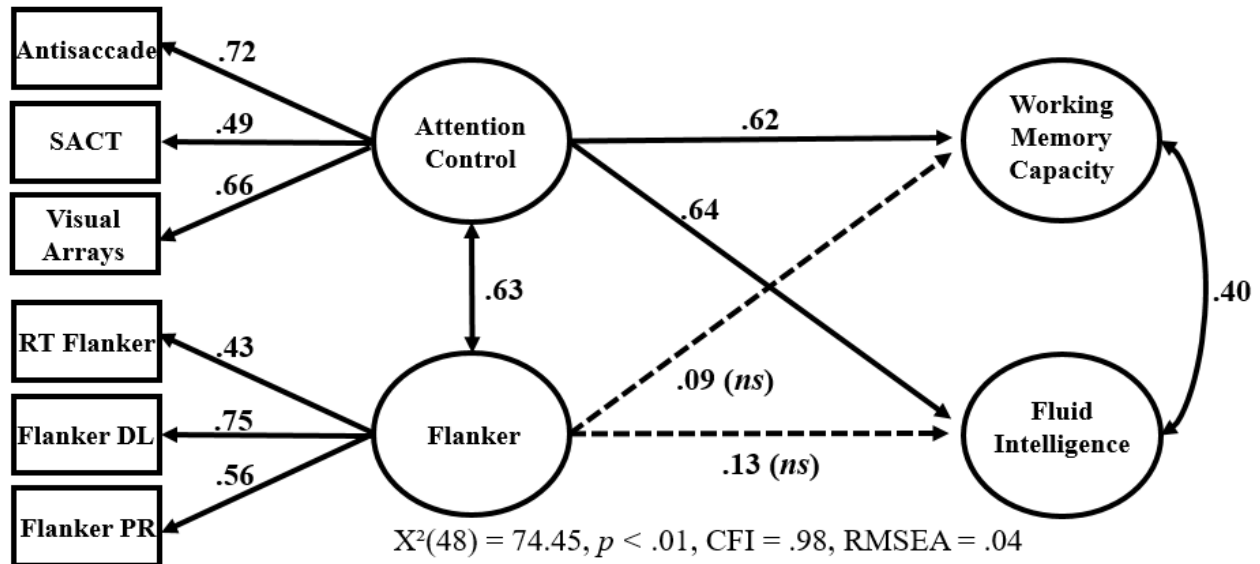


Figure 7 - Structural equation model with attention control and flanker as separate factors predicting working memory capacity and fluid intelligence. Flanker DL = modified flanker with adaptive response deadline; Flanker PR = modified flanker with adaptive presentation rate; SACT = sustained attention-to-cue task. Working memory capacity is comprised of operation span, symmetry span, and rotation span, with standardized loadings of .62, .79, and .80, respectively. The correlation between working memory capacity and fluid intelligence is the correlation between their disturbance terms. Non-significant paths are shown with a dashed line. $N = 383$.

The second question was whether the tasks we have labeled as attention control reflect important variance related to attentive processes, or if instead they are contaminated with construct-irrelevant variance. For instance, individual differences in processing speed or general fluency might explain the high degree of shared variance among the attention tasks, as well as the attention measures to fluid intelligence and working memory capacity (see Hedge et al., 2020). It therefore was necessary to include some measures of discriminant validity to test whether attention control provided unique contribution to criterion measures above and beyond some baseline performance.

We tested this directly using data from a follow-up study in which we administered processing speed tasks to a subset of the same subjects ($N = 173$). The processing speed

measures were computerized versions of existing paper and pencil tasks; letter string comparison, digit string comparison, and digit symbol substitution (these are described in Appendix B, and the full correlation matrix involving these tasks are shown in Table C). We were interested in the extent to which processing speed mediated the relationship between attention control and working memory capacity/fluid intelligence. If, for instance, processing speed accounted for this relationship entirely (full mediation), this would be a strong indication that the attention measures were indeed not process pure and that our results could potentially be attributed to the attention tasks employed here measuring general processing abilities rather than attention control.

We first tested two models with processing speed mediating attention control's relationship to either working memory capacity or fluid intelligence, reported in Appendix C (Figures C1 and C2). These models show that processing speed did not mediate the relationship between attention control and either working memory capacity or fluid intelligence. In Figure 8, we report the critical test of whether general processing speed accounts for why our attention tasks correlated so strongly to working memory capacity and fluid intelligence. Here, processing speed and attention control are correlated but separable factors predicting both working memory capacity and fluid intelligence. If general task fluency or processing speed explained why our attention measures correlated strongly to working memory capacity and fluid intelligence, then we would expect to see the paths from processing speed to working memory capacity/fluid intelligence to be high, and the paths from attention control to working memory capacity/fluid intelligence to be low. In other words, we would see that attention control would not predict much variance in working memory capacity or fluid intelligence *above and beyond* processing speed. Instead, we found the exact opposite. Attention control and processing speed shared

roughly 41% of their variance at the latent level, but processing speed accounted for no incremental variance in working memory capacity/fluid intelligence above and beyond attention control, whereas attention control accounted for substantial variance in these two abilities above and beyond processing speed (69% incremental variance to working memory capacity and 38% to fluid intelligence). The substantial incremental variance contributed by attention control over processing speed shows that processes uniquely captured by the attention tasks are important for explaining the increased relationship to working memory capacity and fluid intelligence, whereas processing speed is not sufficient for account for this relationship.

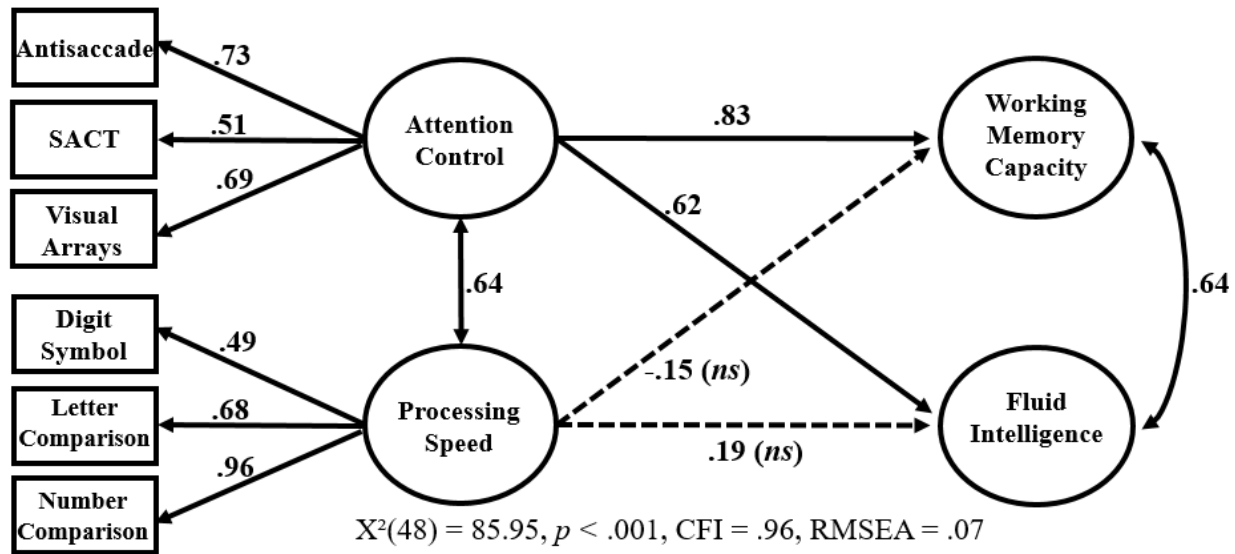


Figure 8 - Structural equation model showing the relative contributions of attention control and processing speed on working memory capacity and fluid intelligence. SACT = Sustained attention-to-cue task. Working memory capacity was comprised of symmetry span, operation span, and rotation span, with loadings of .83, .63, and .81, respectively. Fluid intelligence was comprised of Raven's advanced, number series, and letter sets with loadings of .76, .79, and .84, respectively. The correlation between working memory capacity and fluid intelligence is the correlation between their disturbance terms. Non-significant paths are shown with a dashed line. $N = 173$.

The final set of analyses was conducted to test the possibility that using incongruent or congruent reaction time on the Stroop and flanker tasks as the dependent variable would result in similar improvements as the adaptive Stroop and flanker tasks. In other words, if the main problem with the traditional Stroop and flanker is that they are scored using difference scores, then why not avoid this by just using the component scores (mean reaction time on congruent and/or incongruent trials) instead? There is some precedent to this approach. Kane et al. (2016) reported that residual measures of Stroop and flanker performance (incongruent trials regressed on congruent or neutral) consistently correlated more strongly to within-construct measures than pure reaction time or accuracy difference scores from those tasks. The exception was their number Stroop, which did not correlate with other attention measures when scored either by pure difference score or the residual. For the number Stroop, they therefore used mean reaction time

on incongruent trials and observed some improved correlations to other measures. In another study, Kane and McVay (2012) used incongruent reaction time as the indicator for Stroop performance in their structural equation models over the traditional Stroop effect because, “Stroop incongruent RT showed stronger simple correlations with the other attention-control measures and loaded significantly on an attention-control latent factor” (p. 11).

In the present study, incongruent and congruent reaction times were highly reliable and had stronger correlations to the other executive functioning tasks than did their resulting difference scores. But there is a problem. In Draheim, Mashburn et al. (2019) we warned against the use of mean reaction time on congruent or incongruent trials as the dependent variable in the Stroop and flanker because there is no control for general processing abilities, such as processing speed and task fluency. While contrasts (difference scores) are problematic in correlational research due to unreliability and attenuated correlations, they are used as a control for baseline performance and general processing (though researchers have raised doubts as to the efficacy of difference scores as a control for baseline processing, particular for reaction time differences; e.g., Hedge et al., 2020; Rey-Mermet et al., 2019; Verhaeghen & De Meersman, 1998). Using instead congruent and/or incongruent trial performance in the Stroop and flanker tasks properly accounts for neither speed-accuracy interactions nor general processing. Our data bear this out as well. Reaction time on the congruent and incongruent trials correlated with processing speed quite strongly, an average of $r = .46$, compared to the modified Stroop and flanker tasks or reaction time difference scores from the standard Stroop and flanker (an average of $r = .23$). We tested the statistical significance of the difference between the correlations (Steiger, 1980) and found that both incongruent and congruent reaction time on the Stroop and flanker correlated more strongly to processing speed than their corresponding modified Stroop and flanker variant

($\alpha = .05$; no correction for multiple comparisons). On the other hand, the modified Stroop and flanker tasks did not significantly correlate any stronger to processing speed than the reaction time difference score versions of the tasks. This analysis highlights why using simple mean performance on Stroop and flanker tasks as the dependent variable is not a good approach— these scores correlate more strongly across the board but not discriminately so as there is less control for construct-irrelevant variance. This analysis also suggests that our modified Stroop and flanker tasks were no more contaminated with processing speed than reaction time difference scores from the standard Stroop and flanker tasks.

A limitation is that the above analysis was based on task-level correlations and not latent relationships. We therefore tested a related hypothesis using structural equation modeling (Figure 9) which had three factors predicting attention control (defined here as sustained attention to-cue, visual arrays, and antisaccade). The model was a test of the relative contributions to attention control from the Stroop and flanker tasks with the adaptive response deadline, processing speed, and mean reaction time on Stroop and flanker. That is, how much variance does each contribute when accounting for (partialling out) variance shared with the other factors. In this model, the adaptive response deadline versions of Stroop and flanker used in the present study contributed substantial unique variance (29%) to attention control above and beyond both processing speed and reaction times on congruent/incongruent Stroop and flanker. Further, reaction times on congruent and incongruent Stroop and flanker trials contributed no statistically significant variance to attention control beyond processing speed or the adaptive Stroop and flanker response deadline tasks.

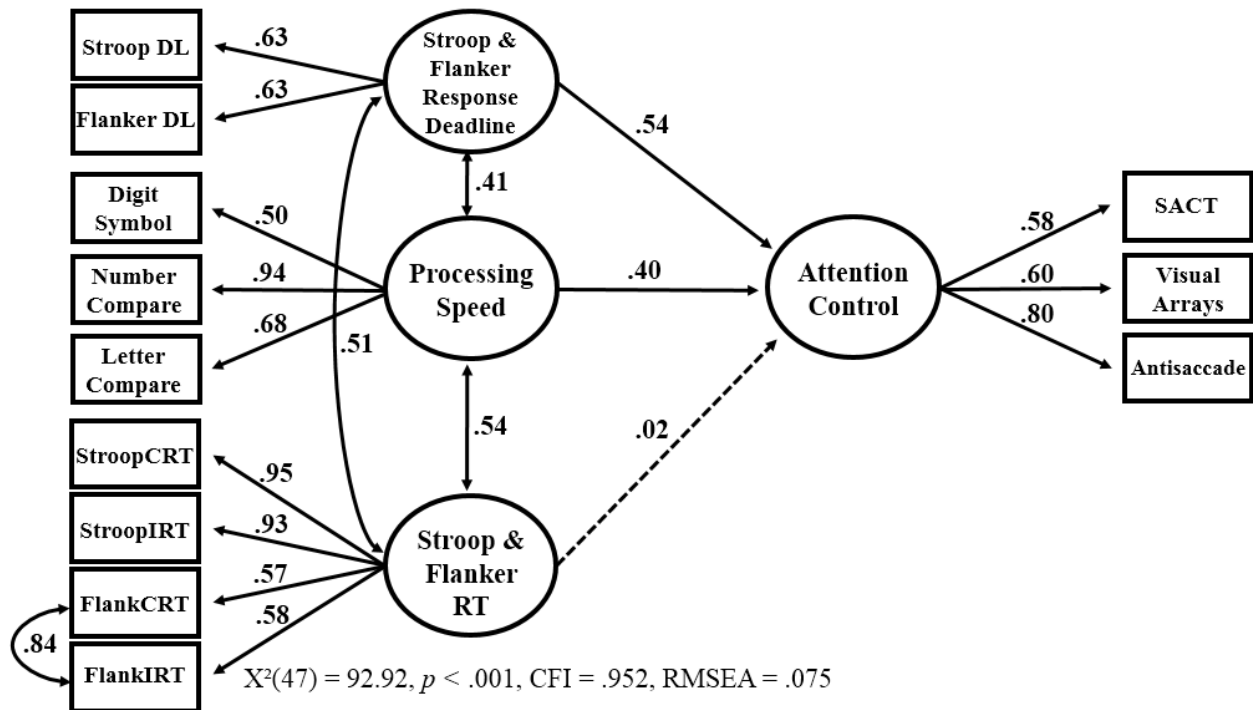


Figure 9 - Structural equation model testing the relative contributions of the Stroop and flanker deadlines tasks, processing speed, and mean reaction time in Stroop and flanker to attention control. StroopCRT = mean reaction time on congruent trials in the standard Stroop task; StroopIRT = mean reaction time on incongruent trials in the standard Stroop task; FlankCRT = mean reaction time on congruent trials in the standard flanker task; FlankIRT = mean reaction time on incongruent trials in the standard flanker task; SACT = Sustained attention-to-cue task. Reaction time on flanker incongruent and congruent trials needed to have correlated error terms for the CFI and RMSEA fit indices to reach acceptable levels, hence why the error terms were correlated for these two indicators. The model without this additional path had a CFI of around .70 and an RMSEA > .10, but all other paths were numerically very similar regardless of whether flanker reaction times were allowed to correlate. Paths are reported as positive if better performance on one task or factor was associated with better performance on another (i.e., a positive path between the adaptive Stroop and flanker response deadlines and attention control indicates that individuals with larger and better scores in the attention tasks had lower response deadline thresholds in the Stroop and flanker deadline tasks). $N = 173$.

The large and positive association between processing speed and attention control even after accounting for the other two factors is also noteworthy. This suggests that processing speed also has independent contributions to attention control that cannot be accounted for by performance in the Stroop or flanker tasks. This would be a cause for concern if performance on the response deadline Stroop and flanker tasks did not also contribute substantial unique variance

to attention control. However, both processing speed and the adaptive response deadline Stroop and flanker tasks contributed unique variance to attention control, which shows that processing speed is at play in tasks such as antisaccade and visual arrays. This is not surprising as we would expect general processing/processing speed to be involved in performance in any executive functioning task to some degree, and it highlights the importance of using structural equation modeling and other regression techniques to partial out variance to test the unique contributions of different abilities.

Discussion

To aid discussion, in Table 7 we report rankings of each task in terms of their performance on the four criteria tested here. On the whole, the accuracy-based measures of attention performed markedly better than the existing reaction time measures. In particular, the reaction time Stroop and flanker measures performed poorly on every criterion. The poor performance of the reaction time difference score Stroop and flanker tasks was expected given the previous literature on their problems (e.g., Rouder & Haaf, 2019).

The antisaccade, visual arrays, modified flanker with adaptive response deadline, and sustained attention-to-cue tasks performed very well in terms of their rank-ordering on all criteria. They were the most stable across two administrations, they were the most strongly inter-correlated, and they best formed a coherent latent factor which in turn had a strong relationship with working memory capacity and fluid intelligence. The selection visual arrays and sustained attention-to-cue tasks correlated to antisaccade at or above $r = .40$. This magnitude of correlation is noteworthy because these are superficially quite different tasks that place different sorts of demands on the respondent. Further, correlations this strong are rarely observed among attention or inhibition measures,

Directly comparing the flanker with adaptive response deadline with the flanker reaction time difference score, the adaptive version improved this task across all criteria. The test-retest reliability, correlational to antisaccade, average factor loading onto an attention factor, and correlation to working memory capacity and fluid intelligence were roughly *doubled* for the adaptive response deadline flanker over the standard reaction time flanker. The exploratory factor analysis and structural equation model presented in Figure 7 shows that the traditional and modified flanker tasks still loaded together, which is one piece of evidence that the manner in which we modified the flanker task improved its psychometrics without drastically changing the nature of the task.

The response deadline Stroop and presentation rate flanker performed in the middle. These measures were an improvement to the traditional reaction time Stroop and flanker tasks, but not as strongly so compared to the antisaccade, visual arrays, sustained attention-to-cue, and response deadline flanker. The Stroop response deadline and flanker presentation rate tasks performed mediocre across the board, though they were still improvements to the reaction time difference score-based Stroop and flanker (Table 7).

The psychomotor vigilance task also performed in the middle but was the best reaction time measure of the three tested. We had reasoned that this might be an example of a reaction time-based measure better suited to individual differences than most other reaction time measures because it is not difference score-based *and* does not appear to be sensitive to speed-accuracy interactions. And indeed the psychomotor vigilance task was highly reliable in terms of internal consistency, had moderately strong inter-correlations with the other attention measures (the strongest of any reaction time measure), and loaded with the other high-performing attention measures in an exploratory factor analysis. However, this task had an average factor loading of

only .40 in the models we tested (Table 5) and had a weak relationship to working memory capacity ($r = .12$). This may suggest that performance on the psychomotor vigilance task reflects processes related to attention control that are less important for performing higher-order cognitive tasks. However, this question is beyond the scope of the present study. The Stroop response deadline and flanker presentation rate tasks performed mediocre across the board, though they were still improvements to the reaction time difference score-based Stroop and flanker (Table 7).

The adaptive visual cue also performed very poorly across the board. One potential reason is that, even after removing outliers that were 3.5 SDs above or below the mean, performance in this task was highly non-normal with skew of 4.91 and kurtosis of 38.39 (see Table 2). However, the results from the adaptive visual cue task did not improve even when we normalized scores using an iterative procedure that continuously removes outliers and calculates a new mean until skewness and kurtosis is under 3. As such, there is a fundamental problem with this task conceptually, in its implementation, and/or in its scoring.

In summary, the adaptive response deadline flanker was an improvement to the standard reaction time flanker; the sustained attention-to-cue with 64 trials was roughly equal in reliability to the psychomotor vigilance task with 80 but an improvement otherwise; and the visual arrays performed on par with the antisaccade.

Table 7

Ranking of attention tasks based on criteria used in present study

Task	Internal Consistency	Test-retest Reliability	Inter-correlation	Factor Coherence	Relationship to WMC and Gf	Rank
Antisaccade	3	1	1	1	2	1
Visual Arrays	4	2	3	2	1	2
SACT	2	4	2	4	6	3
Flanker DL	n/a	6	5	3	3	4
Stroop DL	n/a	3	7	6	4	5
PVT	1	5	4	7	8	6
Flanker PR	n/a	9	5	5	5	6
AVC	n/a	8	8	8	10	8
RT Flanker	6	10	9	8	7	8
RT Stroop	5	7	10	10	9	9

Note. PVT = psychomotor vigilance task; SACT = sustained attention-to-cue; Stroop DL = modified Stroop with adaptive deadline; Flanker PR = modified flanker with adaptive presentation rate; Flanker DL = modified flanker with adaptive response deadline; AVC = adaptive visual cue; WMC = working memory capacity; Gf = fluid intelligence. Values equivalent to the second decimal were considered a tie. Internal consistency was not factored into total rank since it was not calculated for the adaptive tasks. Test-retest rankings reflect reliability after removing outliers for the task. Factor coherence was defined as the average factor loading for each task across all 120 possible tri-indicator models. Relationship to WMC and Gf was defined as the average of each task's correlation to composites of both WMC and Gf.

The mediation analyses (Figure 6) showed that attention control could fully mediate the working memory capacity/fluid intelligence relationship, which is not something we had not observed with previous studies and when using the standard difference score versions of the Stroop and flanker. This full mediation is theoretically interesting because it supports our theory that individual differences in executive functioning arise due to individual differences in executive attention (attention control). However, full mediation of attention control on the

working memory capacity/fluid intelligence relationship was only possible when visual arrays was included in the model – the other new and modified attention tasks did not achieve full mediation without the visual arrays. This is worth exploring further, which we do in Appendix D.

Finally, we found evidence that performance on flanker tasks reflected different variance from other attention measures, although flanker performance did not predict unique variance to working memory capacity or fluid intelligence. This did not appear to occur simply due to having more flanker tasks than, say, Stroop tasks or sustained attention tasks, as we found that flanker performance was separable even when the analyses included only two flanker tasks. These results are consistent with views that flanker tasks reflect different processes than other attention tasks (e.g., Paap et al., 2019). For example, in the Miyake and Friedman (2004) taxonomy, flanker performance relies on the ability to resist distractor interference whereas Stroop and antisaccade require inhibition of a prepotent response. Similarly, in the Kane et al. (2016)'s distinction tasks such as antisaccade and Stroop require restraining attention to override a prepotent response with a novel and goal-directed one, whereas flanker tasks require constrained attention to identify targets among visual distractors. Kornblum's (1994) dimensional-overlap model distinguished two types of conflict in attention tasks, stimulus-stimulus incompatibility or stimulus-response incompatible, with flanker tasks often used to assess the former and Simon tasks the latter (c.f., Paap et al., 2019). Still, the emergence of a coherent and separable flanker factor was surprising. Although flanker performance is thought to reflect different processes than other attention tasks, a number of studies have reported a lack of reliable and valid individual differences variance in standard flanker tasks. For example, Salthouse (2010) concluded that while the flanker task was sensitive to attention-related conflict (and therefore suitable for experimental studies), there were no systematic individual differences

in the resulting flanker scores and so he questioned whether the flanker should even be used as an individual differences measure. Paap et al. (2019) found rather strong correlations among inverse efficiency (reaction time/error rate) scores on a Simon and two Stroop tasks, but not with the flanker task. And Rouder et al. (2019) determined that individual variation in arrow flanker performance from Rey-Mermet et al.'s (2018) data was "so small that it can be accounted for with trial variation alone" (p. 20). This was in contrast to Rey-Mermet et al.'s color Stroop, which had a far greater degree of between-subjects variance relative to trial-level noise.

A commentary on Rey-Mermet et al. (2019)

Our findings are at odds with many individual differences investigations of attention control/inhibition. We found relatively strong inter-correlations among attention control measures, a coherent attention control factor, and strong criterion validity with other cognitive ability measures; other studies of this type generally report the opposite (see Draheim, Mashburn et al., 2019; Friedman & Miyake, 2004; Paap et al., 2019; Rey-Mermet et al., 2018; 2019). Rey-Mermet et al.'s (2019) study is of particular relevance because their goal was putatively the same as ours – to test the unity of attention control and its relationship to working memory capacity/fluid intelligence using accuracy-based attention measures. There are important differences between our study and theirs which likely account for the discrepant results and conclusions. These include their use of difference scores to assess performance and increased attentional demand in their baseline trials due to time pressure and carryover effects. These are discussed in more detail in the following paragraphs.

Rey-Mermet et al. (2019) and the present study used different novel approaches to isolate variance of interest. Here, we included tasks designed to demand greater controlled attention and without requiring scoring via difference scores. We also tested whether improvements in the

tasks were due to the introduction of construct-irrelevant variance common across the tasks such as processing speed and/or general task fluency (Figures 8, C1, and C2). These analyses showed that, while processing speed and attentional control were correlated, attention control predicted a substantial amount of variance in working memory capacity/fluid intelligence over processing speed. Rey-Mermet et al. used a different approach, including a calibration procedure designed to eliminate variance associated with processing speed, episodic memory, and general task ability. However, they did not test the efficacy of their novel manipulations.

One major difference between Rey-Mermet et al. (2019) and the present study was that Rey-Mermet et al. scored all but one of their attention tasks using difference scores. The problems with scoring performance on executive functioning tasks using difference scores were mentioned in the introduction and have been discussed at length elsewhere (e.g., Draheim, Mashburn et al., 2019; Hedge et al., 2018; Hughes et al., 2014; Paap & Sawi, 2016). Difference scores are used despite their known problems because there is a widely held belief that difference scores are necessary to account for baseline performance and therefore can help isolate variance of interest. But there is a strong argument that difference scores are unsatisfactory for this purpose (e.g., Hedge et al., 2020; Verhaeghen & De Meersman, 1998). Rey-Mermet et al. acknowledged this as well in regard to reaction time difference scores:

Subtracting RTs is premised on the assumption of additive factors. That is, the duration of the processes in the baseline condition and the duration of the executive-control process combine additively to the RT in incongruent trials, and the duration of each process is uniquely affected by its own source of individual differences. However, this assumption is questionable because across various speeded tasks, the RTs of slow individuals are related to those of fast individuals through a constant proportional slowing

factor (Zheng, Myerson, & Hale, 2000). This implies that differences between RTs are also proportionally larger for slower than for faster individuals. (p. 1339)

While Rey-Mermet et al. (2019) used accuracy-based difference scores and not reaction time-based ones, it is an open question as to whether accuracy difference scores suffer from the same shortcomings as reaction time differences. We would argue that many of the concerns expressed by Draheim, Mashburn et al. (2019) apply to both reaction time and accuracy difference scores – we took aim at reaction time ones because they are used far more often in assessing executive functioning. Further, Hedge et al. (2020) found that both accuracy and reaction time difference scores in the Stroop, flanker, and Simon tasks are contaminated by processing speed and response cautiousness. Independent of process purity, difference scores are also less reliable than their components, to that end Rey-Mermet et al.'s (2019) measures had an average internal consistency of .70 as opposed to .83 for our non-adaptive measures (see Table 2), although some of their measures were quite reliable. But other methodological factors cannot be ruled out as the reason for the discrepant results, such as differences in population, sample size, number of trials, effect sizes, ratio of baseline trials (congruent or neutral) to incongruent, just to name a few.

Rey-Mermet et al. also employed a within-subject calibration procedure in each of their attention tasks. In this calibration procedure, response deadlines for baseline trials (neutral Stroop and flanker trials, congruent Simon trials, and prosaccade trials) increased or decreased to converge upon a 75% accuracy threshold for each subject. In the experimental blocks which contained the baseline trials plus incongruent trials in the Stroop, flanker, and Simon tasks and pure blocks of antisaccade trials in the antisaccade task. Response deadlines in the experimental blocks were carried over from the calibration procedure. It is not clear how this adaptive

procedure changes the nature of these tasks, and Rey-Mermet et al. (2019) acknowledged some potential pitfalls of this approach. The calibration procedure used by Rey-Mermet et al. may account for their null findings. First, reducing the response deadline of baseline trials (e.g., neutral trials in Stroop, prosaccade in the antisaccade task) during calibration would lead participants to engage in more controlled processing than when not under time pressure. That is, time pressure increases attentional demands. Indeed, studies have shown that task performance is altered under time pressure (Earles, Kersten, Berlin Mas, & Miccio, 2004), that time pressure in Raven's increases its correlation to working memory capacity (Chuderski, 2015), and that the ability to perform under time pressure predicts decision-making performance (Joslyn & Hunt, 1998). Second, Rey-Mermet et al. noted that administering prosaccade trials immediately prior to antisaccade trials could impact antisaccade trials and reduce individual differences (e.g., Kane et al., 2001). Our concern is with the opposite: Kane et al. (2001) showed that there were carryover effects when presenting prosaccade (baseline) trials *after* antisaccade trials such that the prosaccade trials involved more controlled processing, resulting in performance differences between high and low working memory capacity individuals. Rey-Mermet et al. administered calibration (baseline) blocks both before and *after* the subjects performed the critical attention trials. Subjects having performed the experimental trials shortly beforehand would therefore introduce interference into the calibration trials and require the subjects to re-instantiate the new goals of the task (for example, look toward the cue now instead of away from it as you have been previously). We argue that the baseline trials used in Rey-Mermet et al.'s calibration procedure likely involved a significant degree of controlled processing due to these carryover effects and the aforementioned time pressure, therefore reducing not just general construct-irrelevant

variance (such as processing speed), but also reducing the amount of controlled attention reflected in the dependent variables for these tasks.

Alternatively, it is possible that our results and Rey-Mermet et al.'s (2019) are not as contradictory as they appear. We may have different conceptualizations of what these tasks are designed to measure, and we may therefore be simply referring to different abilities despite using shared terminology and tasks. We hold the view that attention control is a broad and domain-general ability responsible in part for individual differences in most, if not all, higher-order cognitive tasks (Engle 2002; 2018; Kane et al., 2007). From this perspective, there may be no separate attention control mechanisms but rather tasks place different demands on attention. Rey-Mermet et al. used the more narrow term “inhibition”, ostensibly conceptualized as a specific cognitive mechanism which is required by tasks such as Stroop, flanker, and Simon. Their argument is that inhibitory (or conflict-resolution) processes in these tasks are task-specific. Perhaps both are true, and that Rey-Mermet et al.'s methodology (difference scores and their calibration technique) reduced, even eliminated, variance associated with general attentional control and therefore the leftover reliable variance reflected the very narrow ability to resolve interference in the specific task at hand, whereas our findings are due to influences of domain-general attention control that cannot be attributable to any single mechanism or ability.

Recommendation for future research

Future research should focus on further development and improvement of attention measures, and to that end the present study provides just one potential method for doing so. For researchers interested in using the tasks employed here, they were programmed in E-Prime and we will make them available for download on our website at <http://englelab.gatech.edu/taskdownloads>.

We can in good conscience recommend the antisaccade and the selective-based visual arrays to other researchers interested in assessing individual differences in attentional control. These are established tasks that display good psychometric properties with relatively little time investment. These tasks were the best performing across all criteria with the exception of internal consistency in the visual arrays, though visual arrays retained most of its reliable variance in test-retest even after an average of around six months between administrations.

Two other promising tasks are the sustained attention-to-cue and the flanker with an adaptive response deadline, with Stroop with an adaptive response deadline just behind those two. These tasks performed quite well, particularly when compared to the traditional reaction time difference score Stroop and flanker tasks. However, these tasks were programmed more as a proof of concept than as finished products. These tasks would very likely benefit from further modification, which we intend to explore in future studies. Researchers interested in using these tasks are encouraged to contact the corresponding author for recommended changes.

We continue to warn against the use of the standard Stroop and flanker tasks in individual differences contexts, as these two tasks expectedly performed the worst among all attention measures. These tasks are known to be problematic (e.g., Rouder & Haaf, 2019). Rouder et al.'s (2019) findings show that typical administrations of these tasks involve a high degree of trial-level noise, and that that hundreds if not more trials may be required to raise Stroop and flanker reliability and correlations to desired levels. Further, the work by Hedge et al. (2020) indicates that these tasks reflect little variance associated with the processes they are intended to measure.

Conclusion

The toolbox approach used here to develop attention control measures appears to be a viable method for moving away from traditional experimental tasks which are not suitable for

correlational pursuits (c.f., Hedge et al., 2018). Specifically, developing new attention tasks which do not rely on difference scores and in a manner that minimizes influences from speed-accuracy interactions and processing speed is a promising method for establishing reliable and valid assessments of attention control. Our results demonstrate that attention control is indeed a unitary concept and, as such, that attention measures can reflect much more than task-specific variance. This finding is contrary to results with difference score-based tasks (e.g., Friedman & Miyake, 2014; Rey-Mermet et al. 2018; 2019; Rouder & Haaf; 2019; Rouder et al., 2019; but see Paap et al., 2019). Although a coherent latent attention factor emerged in the present study, we also found evidence for specific and separable sources of variance in some of tasks (e.g., flanker).

Statistical and post hoc attempts to improve the measurement of attention control had not consistently yielded positive results. It was clear that, as Friedman and Miyake (2004) concluded, new attention control measures were needed to make any further progress. In the present study, we pushed performance variance into accuracy and developed new and modified accuracy-based attention measures designed to measure similar processes as other attention measures but without the methodological problems. The results show that these tasks are improvements to existing attention measures, which has implications for how attention control can be better measured moving forward. To that end we hope to see more development of new attention measures instead of continued reliance on measures that have proven to be inadequate. At the very least, we hope that investigators assessing individual differences in attention control and related processes will be cognizant of the methodological issues with many existing attention tasks and consider taking steps to circumvent these issues. Doing so will improve theoretical

studies involving attention by lowering the likelihood of finding null, misleading, or inconsistent results and thereby reducing the occurrence of misguided conclusions based on these results.

Context of Research

The present research was motivated by the current scientific debate regarding the nature and measurement of attention control. Attention researcher struggle to find a coherent and unified attention control factor, and reliability and validity of commonly used attention measures is generally low. We argue that the reasons for the problems with assessing attention are primarily methodological, not theoretical, and that researchers should not give up on the construct. But several previous statistical attempts to solve the methodological problems with attention control had been unsuccessful. This paper was therefore a test of whether employing new tasks, modified tasks, and tasks not generally recognized as primarily reflecting attention processes could lead to better results.

More broadly, tasks which are excellent for experimental (group differences) purposes are often poorly suited for correlational (individual differences) purposes. This phenomenon is beginning to receive more attention in the literature due to the increased awareness of reliability and validity issues with a number of ubiquitous cognitive tasks. The hope is that our line of work will raise awareness to these issues as they pertain to attention control specifically but also in the broad sense. And while there are a variety of ways to address the problem, the authors primarily aim to solve it through continued modification and development of tasks to have more desirable psychometric properties.

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Appendix A

Correlation matrix for attention measures

Table A

Zero-order inter-correlations among the attention control measures

Task	Threshold DV				Accuracy DV			Reaction time DV		
	1	2	3	4	5	6	7	8	9	10
1. Flanker DL	-									
2. Flanker PR	.38*	-								
3. Stroop DL	.26*	.30*	-							
4. AVC	.13*	.06	.08	-						
5. Antisaccade	.34*	.25*	.31*	.33*	-					
6. Visual Arrays	.18*	.27*	.20*	.20*	.45*	-				
7. SACT	.25*	.27*	.29*	.27*	.40*	.31*	-			
8. PVT	.20*	.20*	.21*	.20*	.35*	.25*	.42*	-		
9. RT Stroop	.10	.09	.23*	.09	.19*	.18*	.01	.01	-	
10. RT Flanker	.33*	.21*	.04	.04	.16*	.15*	.13*	.10*	.17*	-

Note. Pairwise-deletion method was used for missing values, correlations are based on a range of $N = 382 - 390$. PVT = psychomotor vigilance task; SACT = sustained attention-to-cue; RT Stroop and RT flanker are reaction time difference scores (Stroop and flanker effect, respectively); Stroop DL = modified Stroop with adaptive deadline; Flanker PR = modified flanker with adaptive presentation rate; Flanker DL = modified flanker with adaptive response deadline; AVC = adaptive visual cue. For ease of interpretation, a positive correlation indicates better performance on both tasks.

Appendix B.

Description of processing speed tasks

All processing speed measures were computerized versions of paper and pencil tests. In each case, subjects were instructed to respond as quickly and accurately as possible, but consistent with standard administration procedures, were not alerted of the time limits of each task in the instruction phase.

Letter String Comparison (Conway, Cowan, Bunting, Theriault, & Minkoff, 2002). In this version of the Letter String Comparison Task, subjects viewed strings of three, six, or nine consonants appearing to the left and right of a central line. The letter strings could either be the same or differ by a single letter. If different, the mismatching letter could appear in any location in the string. Subjects indicated their response by clicking on a button on the screen labeled SAME for identical strings or DIFF for mismatching strings. Letters were printed in white size 18 Courier New font on a black background. After completing six practice trials, subjects completed two 30-second blocks of the task. The dependent variable was the number of accurate responses across both blocks.

Digit String Comparison. The Digit String Comparison task was identical to the Letter String Comparison, except that subjects viewed and made judgements about strings containing three, six, or nine digits.

Digit Symbol Substitution (Wechsler, 1991). This adaptation of the Digit Symbol Substitution Task has been modified to make it more amenable to computer administration and response collection via a standard number pad. The symbols used are the same as the paper and pencil version of the task, and we endeavored to maintain the same basic demands. However, rather than viewing digits and reporting corresponding symbols, this task required subjects to

view symbols and report the corresponding digits. On each trial, subjects were presented with two boxes stacked one on top of the other in the center of the screen. A symbol appeared in the bottom box. Subjects were to consult a key presented at the top of the screen, and to indicate via key press with the digit that belonged in the top box. After ten practice trials, subjects completed 90 seconds of the task. The dependent variable was the number of correctly reported digits during that 90 second period.

Appendix C

Additional processing speed analyses

Table C. Full correlation matrix for analyses with processing speed.

	RAPM	NumSeries	LetterSets	OSpan	SymSpan	RotSpan	Antisaccade	Flanker	FlankerDL	FlankerPR	Stroop	StroopDL	VA4	PVT	SACT	AVC	LetterComp	NumComp	DigitSymb
RAPM		0.520***	0.569***	0.435***	0.519***	0.506***	0.271**	-0.155	-0.247**	-0.126	-0.253**	-0.205*	0.316***	-0.101	0.124	-0.018	0.187*	0.272**	0.446***
NumSeries	0.520***		0.615***	0.385***	0.443***	0.482***	0.261**	-0.102	-0.201*	-0.046	-0.127	-0.077	0.282**	-0.020	0.078	0.147	0.178*	0.284**	0.279**
LetterSets	0.569***	0.615***		0.374***	0.499***	0.462***	0.292***	-0.170	-0.188*	-0.015	-0.231**	-0.083	0.310***	-0.117	0.245**	0.109	0.396***	0.444***	0.450***
OSpan	0.435***	0.385***	0.374***		0.537***	0.411***	0.199*	-0.085	-0.144	-0.047	-0.061	-0.256**	0.222*	0.102	0.049	0.104	0.105	0.123	0.216*
SymSpan	0.519***	0.443***	0.499***	0.537***		0.670***	0.370***	-0.234**	-0.296***	-0.264**	-0.217*	-0.255**	0.488***	-0.048	0.245**	0.056	0.257**	0.285**	0.357***
RotSpan	0.506***	0.482***	0.462***	0.411***	0.670***		0.359***	-0.233**	-0.327***	-0.277**	-0.253**	-0.159	0.402***	-0.151	0.118	0.104	0.106	0.169	0.266**
Antisaccade	0.271**	0.261**	0.292***	0.199*	0.370***	0.359***		-0.179*	-0.242**	-0.263**	-0.219*	-0.310***	0.390***	-0.202*	0.337***	-0.289***	0.281**	0.397***	0.293***
Flanker	-0.155	-0.102	-0.170	-0.085	-0.234**	-0.233**	-0.179*		0.450***	0.183*	0.270**	0.020	-0.104	-0.034	-0.134	0.101	-0.196*	-0.217*	-0.146
FlankerDL	-0.247**	-0.201*	-0.188*	-0.144	-0.296***	-0.327***	-0.242**	0.450***		0.449***	0.156	0.280**	-0.075	0.069	-0.155	0.052	-0.125	-0.155	-0.250**
FlankerPR	-0.126	-0.046	-0.015	-0.047	-0.264**	-0.277**	-0.263**	0.183*	0.449***		0.119	0.391***	-0.174*	0.294***	-0.235**	0.035	-0.025	-0.056	-0.127
Stroop	-0.253**	-0.127	-0.231**	-0.061	-0.217*	-0.253**	-0.219*	0.270**	0.156	0.119		0.210*	-0.136	0.003	-0.125	0.038	-0.180*	-0.250**	-0.175*
StroopDL	-0.205*	-0.077	-0.083	-0.256**	-0.255**	-0.159	-0.310***	0.020	0.280**	0.391***	0.210*		-0.153	0.092	-0.292***	0.029	-0.086	-0.156	-0.178*
VA4	0.316***	0.282**	0.310***	0.222*	0.488***	0.402***	0.390***	-0.104	-0.075	-0.174*	-0.136	-0.153		-0.252**	0.233**	-0.234**	0.315***	0.384***	0.352***
PVT	-0.101	-0.020	-0.117	0.102	-0.048	-0.151	-0.202*	-0.034	0.069	0.294***	0.003	0.092	-0.252**		-0.414***	0.268**	-0.198*	-0.176*	-0.151
SACT	0.124	0.078	0.245**	0.049	0.245**	0.118	0.337***	-0.134	-0.155	-0.235**	-0.125	-0.292***	0.233**	-0.414***		-0.278**	0.284**	0.246**	0.190*
AVC	-0.018	0.147	0.109	0.104	0.056	0.104	-0.289***	0.101	0.052	0.035	0.038	0.029	-0.234**	0.268**	-0.278**		-0.134	-0.145	0.038
LetterComp	0.187*	0.178*	0.396***	0.105	0.257**	0.106	0.281**	-0.196*	-0.125	-0.025	-0.180*	-0.086	0.315***	-0.198*	0.284**	-0.134		0.717***	0.255**
NumComp	0.272**	0.284**	0.444***	0.123	0.285**	0.169	0.397***	-0.217*	-0.155	-0.056	-0.250**	-0.156	0.384***	-0.176*	0.246**	-0.145	0.717***		0.368***
DigitSymb	0.446***	0.279**	0.450***	0.216*	0.357***	0.266**	0.293***	-0.146	-0.250**	-0.127	-0.175*	-0.178*	0.352***	-0.151	0.190*	0.038	0.255**	0.368***	

Computed correlation used pearson-method with listwise-deletion

We tested whether processing speed mediated the relationship between attention control and either working memory capacity or fluid intelligence separately. The attention factor consisted of the antisaccade, visual arrays, and sustained attention-to-cue. In both these models (Figure C1 and C2), attention control predicted a substantial amount of variance in working memory capacity (64%) and fluid intelligence (33%) above and beyond processing speed. Processing speed's paths to working memory capacity and fluid intelligence were also not significant when accounting for attention control, indicating that processing speed did not at all mediate the relationship between attention control and working memory capacity or fluid intelligence.

Because previous results indicated that the visual arrays had a strong relationship with working memory capacity and fluid intelligence, we wanted to ensure that this particular task was not unduly influencing the results. So, we tested models using the modified flanker with adaptive response deadline in place of the visual arrays, and the results were similar.

Figure C1

Processing speed mediating the relationship between attention control and working memory capacity

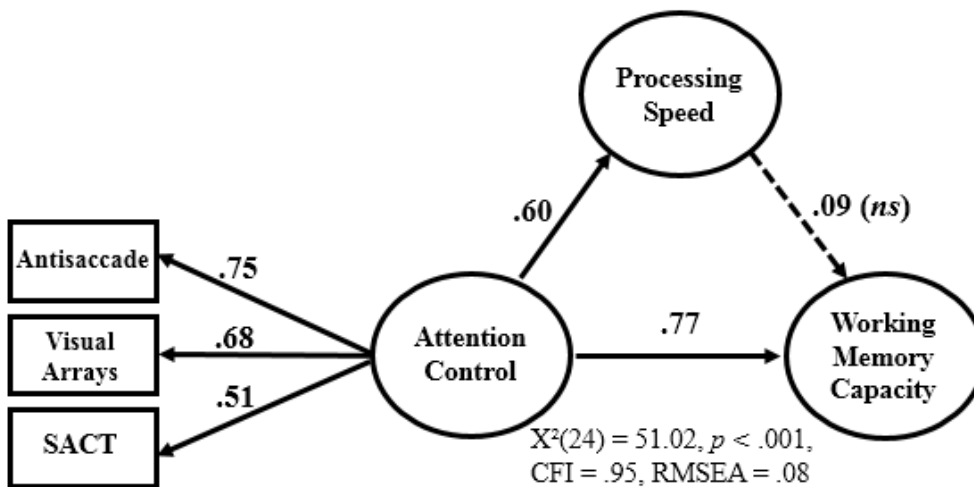
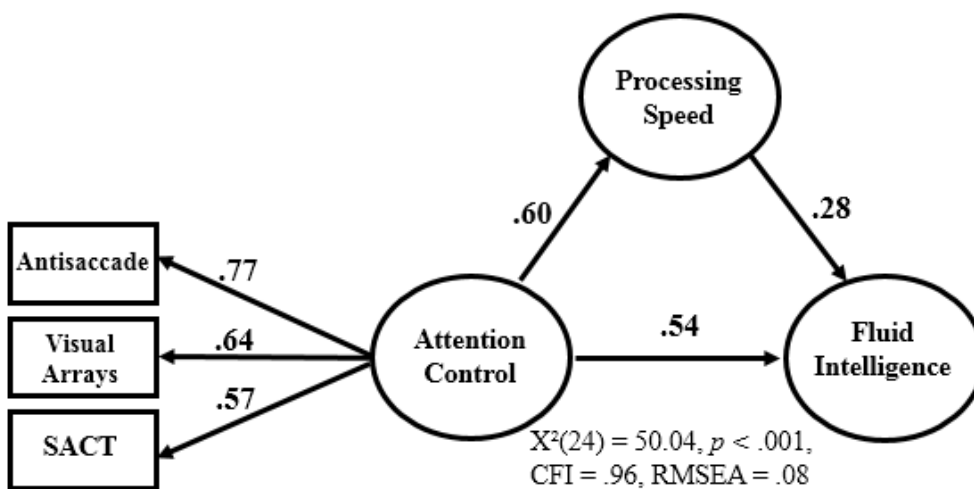


Figure C2

Processing speed mediating the relationship between attention control and fluid intelligence



Appendix D

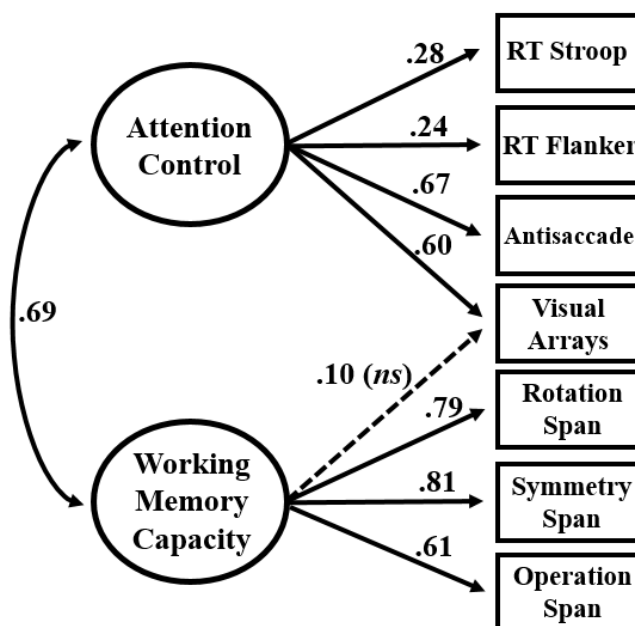
The visual arrays was required to fully mediate the working memory capacity/fluid intelligence relationship

Although attention control fully mediated the working memory capacity/fluid intelligence relationship, this is only true when visual arrays was included as one of the indicators of attention. The modified and new attention tasks did not achieve full mediation alone, or even with the antisaccade. There are a number of potential and non-mutually exclusive explanations for this which are explored in further detail below.

One possibility is that the full mediation occurs with selection visual arrays because the visual arrays is in fact a measure of visual working memory capacity and not attention. If so, then the full mediation is simply due to misspecification and is not theoretically meaningful. We cover the extensive lines of evidence for the visual arrays as a measure of attention in Martin, Tsukahara et al. (2019), but one simple test of whether the visual arrays loads more with attention control or working memory capacity is to compare the visual arrays factor loadings in a model in which it is cross-loaded onto both working memory capacity and attention control. This model is shown in Figure D1, and the results are quite clear – the selective visual arrays used in this study loaded with attention control (namely antisaccade) strongly and significantly (.60) and did not load with working memory capacity (.10, non-significant). This rather straightforward model illustrates that the specific visual arrays task we used (with a selection demand) is better conceptualized as an attention task rather than a working memory task.¹¹ Relatedly, if the full mediation of attention control on the working memory capacity/fluid intelligence relationship required visual arrays because selective visual arrays is a working memory task, then we would

¹¹ In Martin, Tsukahara et al. (2019) we found that these results hold when working memory capacity is measured by tasks other than complex span as well which is replicated over three other large-scale data sets.

see a unique contribution of variance from visual arrays to working memory capacity when the variance common to the attention tasks was partialled out. This model is shown in Figure D2. The selective visual arrays shared 44% of its variance with the attention control factor but it did *not* predict any statistically significant variance in working memory capacity above and beyond attention control and, conversely, these attention measures contributed strong variance (38%) to working memory capacity above and beyond visual arrays. As such, it does not appear that the mediation of attention control on the working memory capacity/fluid intelligence relationship when the visual arrays is included as a measure of attention control is due to misspecification or redundancy of indicators (i.e., using a working memory measure to mediate a relationship involving working memory capacity).



$$\chi^2(12) = 27.78, p < .006, CFI = .97, RMSEA = .03$$

Figure D1 – Confirmatory factor analysis with visual arrays crossloaded onto WMC and AC. RT Stroop and RT flanker are the mean differences in reaction time on congruent and incongruent trials. Paths are reported as positive if better performance on one task or factor was associated with better performance on. The factor loadings for the RT Stroop and RT flanker are unacceptably poor, however they were used as indicators of attention control here for the sake of demonstration and because this is a more standard factor of attention control. $N = 396$.

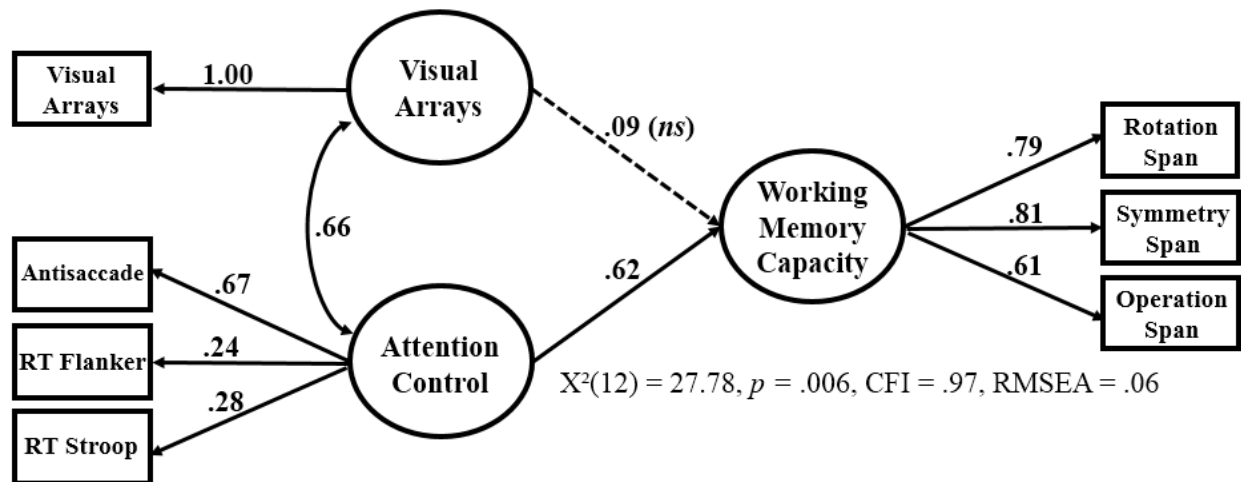


Figure D2 – Structural equation model testing whether visual arrays contributes unique variance to working memory capacity above and beyond attention control. RT Stroop and RT flanker are the mean differences in reaction time on congruent and incongruent trials. Paths are reported as positive if better performance on one task or factor was associated with better performance on. The factor loadings for the RT Stroop and RT flanker are unacceptably poor, however they were used as indicators of attention control here for the sake of demonstration and because this is a more traditional factor of attention control. $N = 396$.

One speculative explanation is that the full mediation of the working memory capacity/fluid intelligence relationship requiring visual arrays was not due to specific processes in the visual arrays, but rather is due to visual arrays improving the attention factor as a whole. The question is, is there something special about the visual arrays itself? Or perhaps the selective visual arrays results in a stronger attention factor, as a whole, due to the manifestation of more theoretically relevant shared variance across the attention tasks. Further research is required to better explore this idea, but we did test one additional model in which the selective visual arrays was the sole indicator of attention control and mediated the working memory capacity/fluid intelligence relationship. The logic here was that if there was something special about the visual arrays by itself then the visual arrays should mediate the working memory capacity/fluid intelligence relationship completely (or near completely) by itself. The model (Figure D3) shows that the selective visual arrays task is a weak mediator of the working memory capacity/fluid

intelligence relationship on its own (the direct path between working memory capacity and fluid intelligence is .58), only achieving a partial mediation. We replicated these results in a reanalysis of Shipstead et al. (2014) in which we used two selective visual arrays tasks as the mediator of the working memory capacity/fluid intelligence relationship with working memory capacity measured using two complex span and two running span tasks - the direct path between working memory capacity and fluid intelligence was .70 after accounting for the mediation of the selective visual arrays tasks. This finding speaks not only to the possibility that the selective visual arrays is improving the attention factor in previous models, but again that visual arrays is not the sole cause of the full mediation of attention control onto the working memory capacity/fluid intelligence relationship. The full mediation was not due to any single task individually but rather was due to the common variance across the tasks.

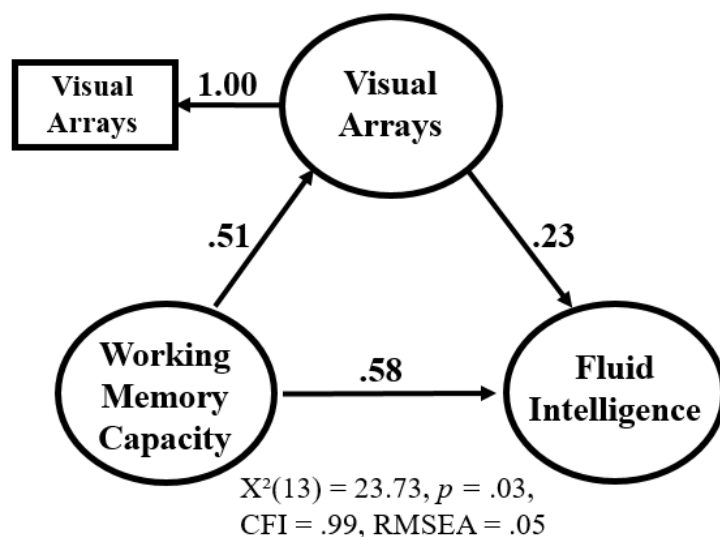


Figure D3 – Structural equation model testing whether visual arrays solely mediates the working memory capacity/fluid intelligence relationship. Working memory capacity was comprised of operation span, symmetry span, and rotation span with respective loadings of .62, .80, and .79; fluid intelligence was comprised of Raven, number series, and letter sets with respective loadings of .75, .72, and .73. $N = 396$.